INTERNET CENSORSHIP
PART II: INFORMATION HIDING

CH. 4: INTRODUCTION TO NETWORK INFORMATION HIDING



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Definition

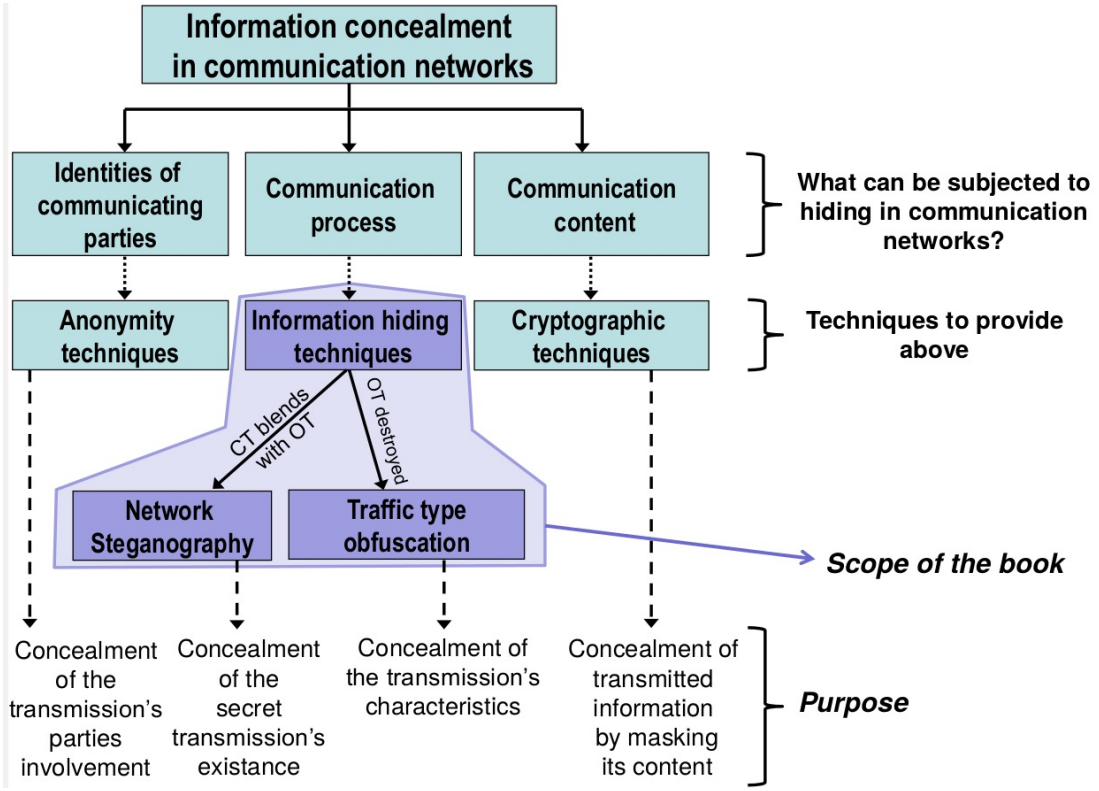


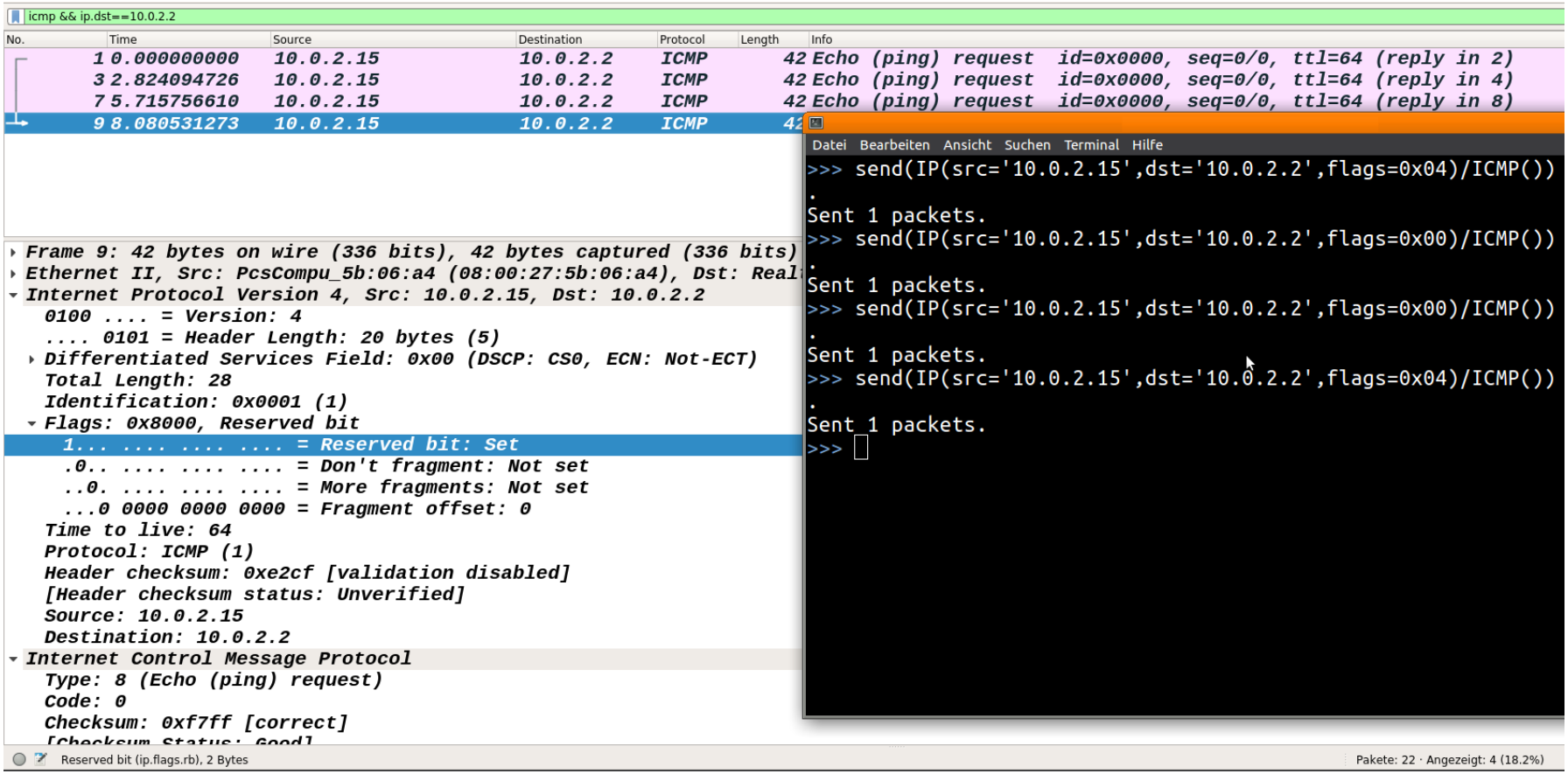
Fig.: W. Mazurczyk, S. Wendzel, S. Zander et al.: Information Hiding in Communication Networks, Wiley-IEEE, 2016

Differences to traditional digital media steganography

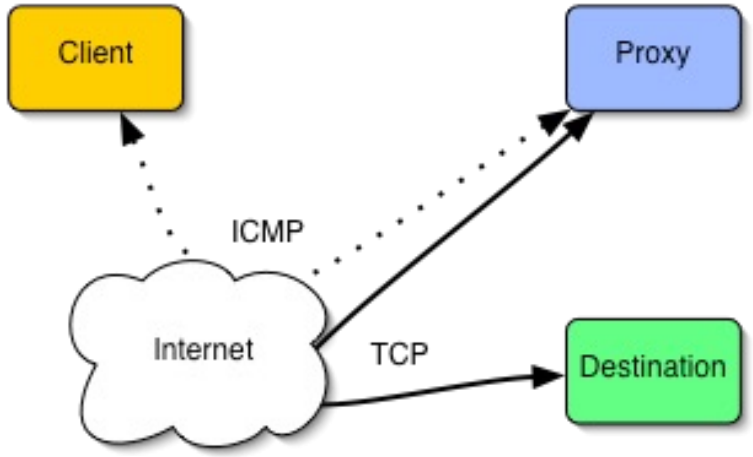
- **Inconsistent terminology**: no clear distinction between **steganography** and **covert channel**
  - See Ch. 1 for definitions of the terms steganography and covert channel and that both are considered as different research domains (covert channels in MLS context!).
  - Thus, in the network context: **network covert channel** or **network steganographic channel** handled **separately**
  - Unified: a steganographic **method** creates such a **covert channel** [1, Chapter 3]
- A bit more terminology:
  - Covert data is hidden in *overt* network transmissions
  - The „cover object“ is now called „carrier“ in the network context
  - Advantage of a constant transmission (e.g. permanent data leakage)
- Advantages:
  - Difficult to analyze **all** network data; smaller delay; with the growth of the Internet, the options for network IH grew and grow, too.

[1] W. Mazurczyk, S. Wendzel, S. Zander et al.: Information Hiding in Communication Networks, Wiley-IEEE, 2016

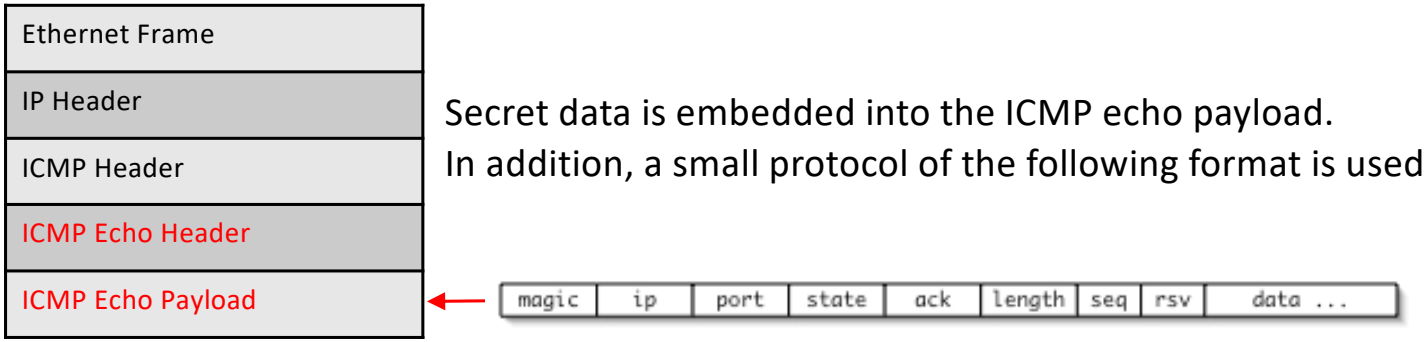
Example 1: Trivial Network Covert Channel via IPv4 Reserved Bit, sending message ``1001”



Example 2: Ping Tunnel



Analysis and improvements:
Jaspreet Kaur, Steffen Wendzel, Omar Eissa, Jernej Tonejc, Michael Meier: Covert Channel-internal Control Protocols: Attacks and Defense, *Security and Communication Networks (SCN)*, Vol. 9(15), Wiley, 2016.



Secret data is embedded into the ICMP echo payload.
In addition, a small protocol of the following format is used:

Figs.: http://www.cs.uit.no/%7Edaniels/PingTunnel/

Types of (Network) Covert Channels

- **Fundamental:**
  - **Local** and **network** covert channels
  - **Storage** and **timing** channels
  - **Noisy** and **noise-free** covert channels
  - **Direct** and **indirect** covert channels
  - **Intentional** and **unintentional** covert channels

Types of (Network) Covert Channels: Storage/Timing Cov. Channels

- Modulate **stored** information or modulate **timing** behavior

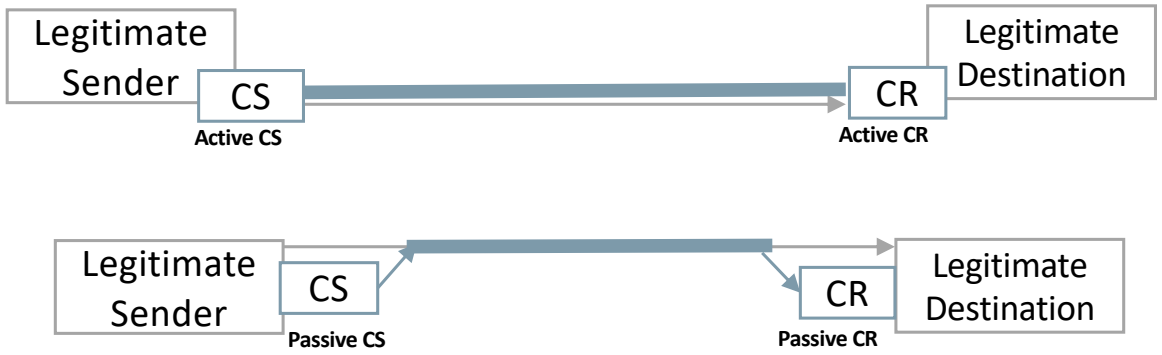
- Storage and timing are the two major categories for categorizing storage channels.

- Flawed, as we have shown in [1].
  - Example: some covert channels modify the order of network packets, such as TCP segments, which would be both, a timing operation (when to send which packet) as well as a storage operation (sequence numbers in packet headers must be written and are also interpreted by the covert channel).

[1] S. Wendzel et al.: A Generic Taxonomy for Steganography Methods, ACM Computing Surveys, 2025.

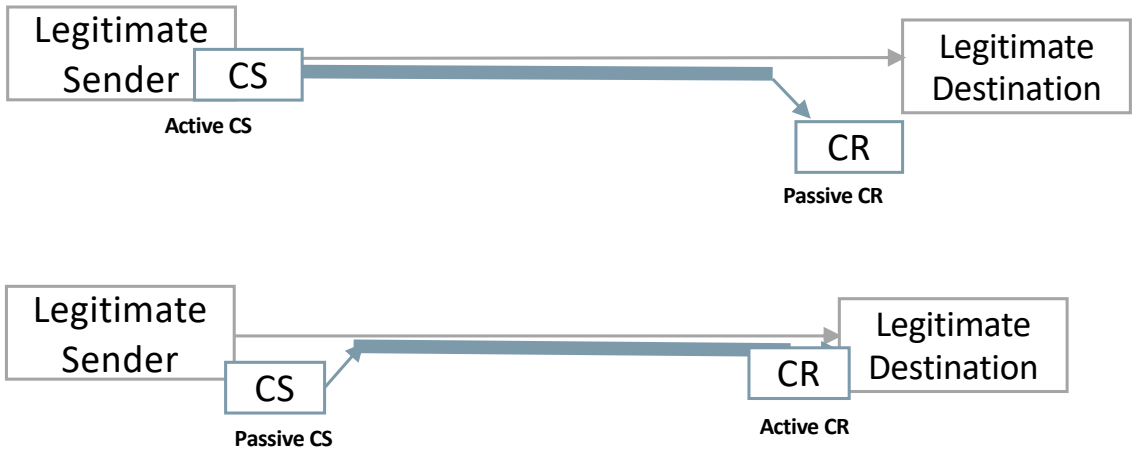
Types of (Network) Covert Channels: Active/Passive Cov. Channels

- **Active** and **passive** Covert Channels (passive elements have a different sender/receiver than the legitimate sender/receiver)



Types of (Network) Covert Channels: Semi-active/passive Cov. Channels

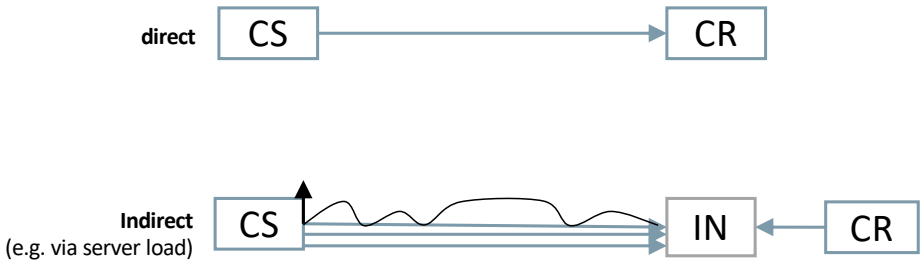
- **Semi-active** and **semi-passive** Covert Channels [1]
  - **Semi-active:**
  - **Semi-passive:**



[1] K. Lamshöft, J. Dittmann: *Assessment of Hidden Channel Attacks: Targetting Modbus/TCP*, IFAC-PapersOnLine, 53(2), 2020.

Types of (Network) Covert Channels: (In)Direct Cov. Channels

- **Direct** and **indirect** covert channels: direct channels do not rely on intermediate nodes (IN).
  - Example: via web page + server load
  - General illustration:



Further differentiation into two major patterns for the intermediate node (IN): **redirector** and **broker**.

- A broker can be a **proxy** or a **dead drop**.

⇒ **Survey:** T. Schmidbauer, S. Wendzel: *SoK A Survey of indirect network-level covert channels*, in Proc. 17th AsiaCCS, ACM, 2022. https://doi.org/10.1145/3488932.3517418

⇒ **Lets have a look into the paper and understand the patterns (broker (proxy; dead drop) and redirector.**

Types of (Network) Covert Channels: (In)Direct Cov. Channels – Case of Dead Drops with ARP [1]

- In general, **network** steganography can only be used to **transfer** secret data.
  - There are a few exceptions such as the Dead Drop (see Ch. 4).

- Illustration of the Dead Drop introduced in [1, text below is a copy from the paper]. Approach with ARP+SNMP (see Fig.):
  a) CS possesses secret information that it wants to store in the network-accessible ARP cache.
  b) CS exploits the ARP cache of a third-party system by sending a fake ARP request containing a (MAC,IP) tuple. The actual host reflected by the (MAC,IP) tuple does not exist and represents encoded secret information.
  c) The third-party host Dead Drop (D) adds the tuple to its local ARP cache …
  d) … where it can be requested by the CR for some time depending on the lifetime of ARP cache entries on the third-party host.

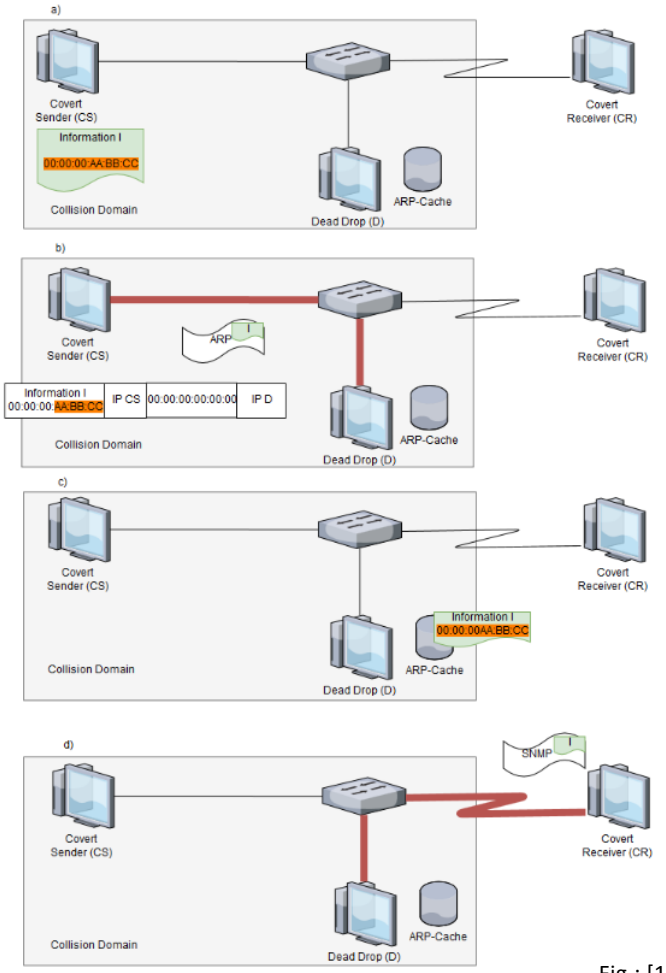


Fig.: [1]

[1] T. Schmidbauer, S. Wendzel, A. Mileva, W. Mazurczyk: Introducing Dead Drops to Network Steganography using ARP-Caches and SNMP-Walks, in Proc. ARES 2019, ACM.

Types of (Network) Covert Channels: (In)Direct Cov. Channels – Case of Dead Drops with ARP [1]

- Lifetime of entries depends on the system and caching limits (all Figures taken from [1]):

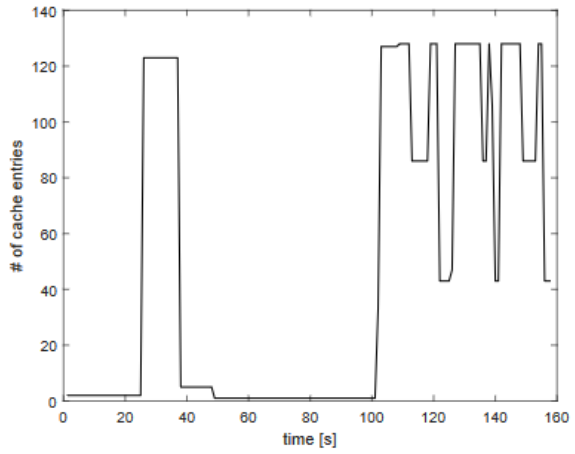


Figure 5: OpenSuSE caching behavior

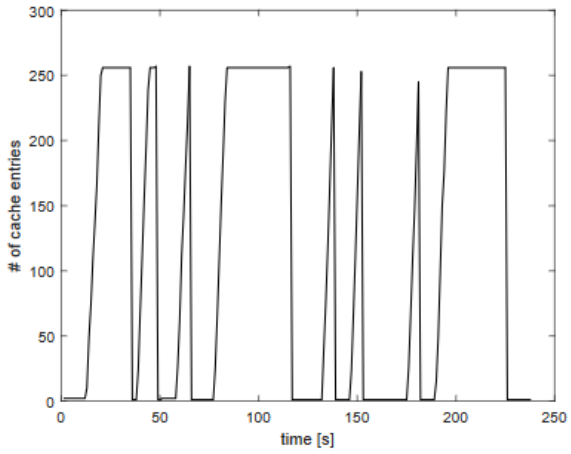


Figure 7: Windows 7 caching behavior

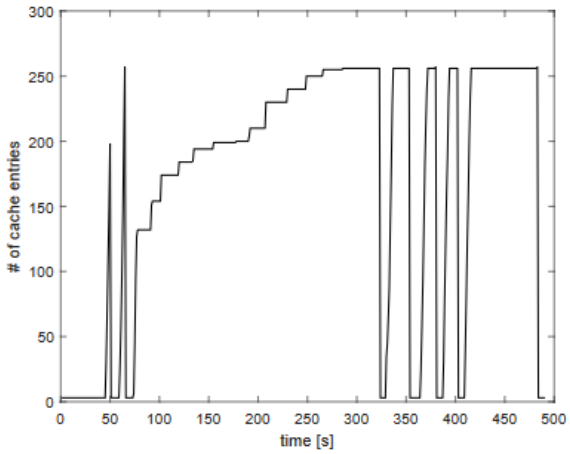


Figure 6: Windows 10 caching behavior

Figs.: [1]

[1] T. Schmidbauer, S. Wendzel, A. Mileva, W. Mazurczyk: Introducing Dead Drops to Network Steganography using ARP-Caches and SNMP-Walks, in Proc. ARES 2019, ACM.

Types of (Network) Covert Channels: (In)Direct Cov. Channels – Case of Dead Drops

- **Advantage** of Network Dead Drops: CS and CR can write/read the secret information at different times, raising fewer attention for their action.

- **Note:** Another approach is available (see a 2020-paper by Schmidbauer et al. that shows the exploitation of the NTP protocol for the purpose of creating a Dead Drop).

Types of (Network) Covert Channels: (Un)Intentional Cov. Channels

- **Intentional** (**covert**) and **unintentional** (**side**) channels
  - e.g. side channels in web applications [1]

  - Example:



Usually, traffic needs to be sent many times and measured exactly to gain any useful information out of this.

[1] S. Schinzel: An efficient mitigation method for timing side channels on the Web, in Proc. 2nd Int. Workshop on Constructive Side-Channel Analysis and Secure Design (COSADE), 2011.

Types of (Network) Covert Channels: Cover Types

Zander categorizes covert channel carriers as follows:

- **Predictable**
- **Variable**
- **Random**

S. Zander: *Performance of Selected Noisy Covert Channels and Their Countermeasures in IP Networks*, PhD Thesis, Swinburn University of Technology, Melbourne, 2010.



History Covert Channels & Covert Channel Amplification

(DYST published: TDSC, 2025)

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Types of (Network) Covert Channels: History Cov. Channels and Cov. Ch. Amplification [1]

- Most covert channels focus on the **present**, e.g., packets might contain secret stego data in their **current** payload.

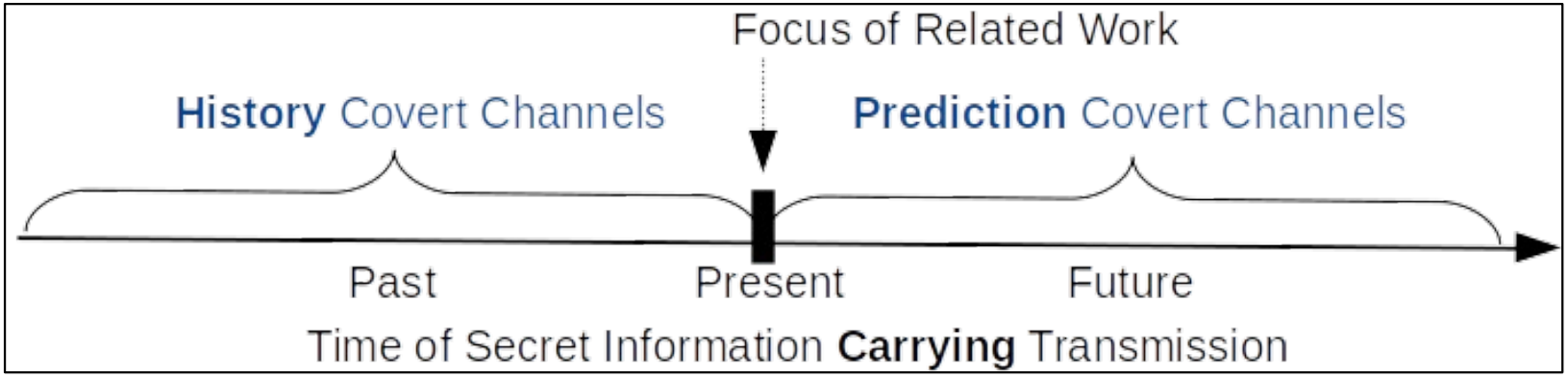
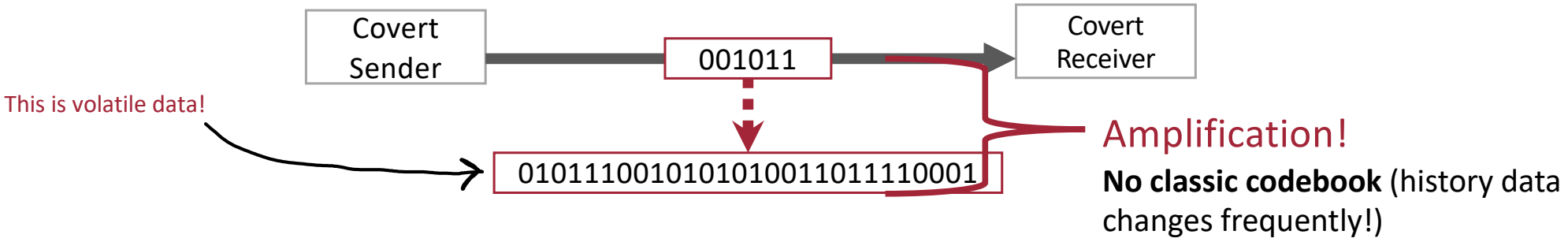


Fig.: [1]

[1] S. Wendzel, T. Schmidbauer, S. Zillien, J. Keller: DYST (Did You See That?): An Amplified Covert Channel That Points To Previously Seen Data. IEEE Transactions on Dependable and Secure Computing (TDSC), Vol. 22(1), 2025, DOI: 10.1109/TDSC.2024.3410679

Types of (Network) Covert Channels: History Cov. Channels and Cov. Ch. Amplification [1]

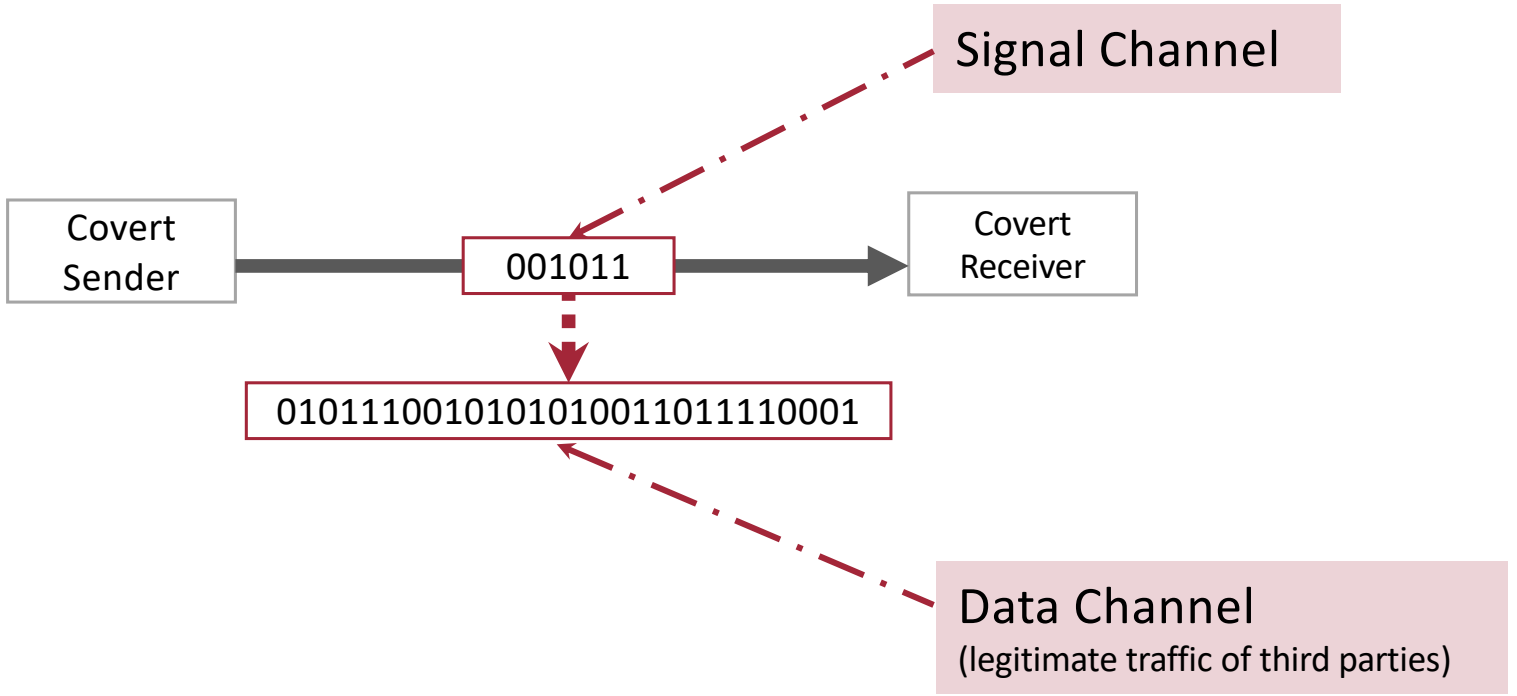
- **History** covert channels optimize transmission sizes by transferring **solely pointers to larger data chunks already seen somewhere**, e.g., in previous packets or online data [1].
- These data chunks represent the actual secret information. First implementation called "**DYST**" (*Did You See That?*). Since the pointers are smaller than the actual secret data, one achieves an **amplification**.



[1] S. Wendzel, T. Schmidbauer, S. Zillien, J. Keller: DYST (Did You See That?): An Amplified Covert Channel That Points To Previously Seen Data. IEEE Transactions on Dependable and Secure Computing (TDSC), Vol. 22(1), 2025, DOI: 10.1109/TDSC.2024.3410679

Types of (Network) Covert Channels: History
Cov. Channels and Cov. Ch. Amplification [1]

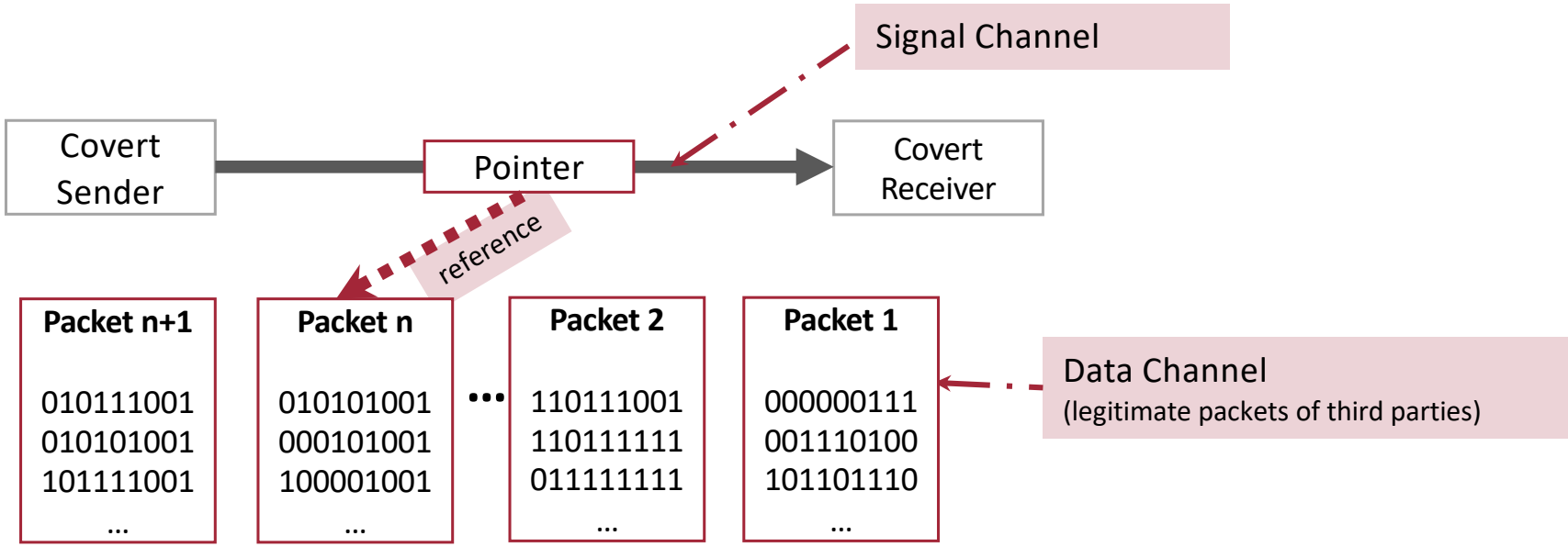
- We split our covert channel into a **signal channel** and a **data channel**.



[1] S. Wendzel, T. Schmidbauer, S. Zillien, J. Keller: DYST (Did You See That?): An Amplified Covert Channel That Points To Previously Seen Data. IEEE Transactions on Dependable and Secure Computing (TDSC), Vol. 22(1), 2025, DOI: 10.1109/TDSC.2024.3410679

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Types of (Network) Covert Channels: History
Cov. Channels and Cov. Ch. Amplification

- History/prediction channels enable a new category of **fully-passive covert channels** (see **figure** below), where a stego **data channel** (in this case "**DYST**") can be represented through 100% legitimate traffic – solely the **signaling channel** (containing the pointer) needs to craft new/modify existing packets [1].

		Covert **Sender**		
		Active (generates own overt traffic in which it embeds covert data)	**Passive** (embeds covert data in overt traffic of third-party nodes)	**Fully-passive** (utilizes third-party traffic without modifying it)
Covert **Receiver**	**Active** (is the destination of the overt traffic)	**Active** Covert Channel	**Semi-passive** Covert Channel	**Fully-and-semi-passive** Covert Channel
	Passive (is not the direct destination of the overt traffic, e.g., a router)	**Semi-active** Covert Channel  **DYST's Signal Channel**	**Passive** Covert Channel	**Fully-passive** Covert Channel  **DYST's Data Channel**

Fig.: [1]

[1] S. Wendzel, T. Schmidbauer, S. Zillien, J. Keller: DYST (Did You See That?): An Amplified Covert Channel That Points To Previously Seen Data. IEEE Transactions on Dependable and Secure Computing (TDSC), Vol. 22(1), 2025, DOI: 10.1109/TDSC.2024.3410679

Types of (Network) Covert Channels: History
Cov. Channels: DYST (Basic Version)

- How do history covert channels work?
  - Different approaches feasible, also outside of networks.
- Together with the concept of history covert channels, we introduced a first implementation (before-mentioned **DYST**) in [1].

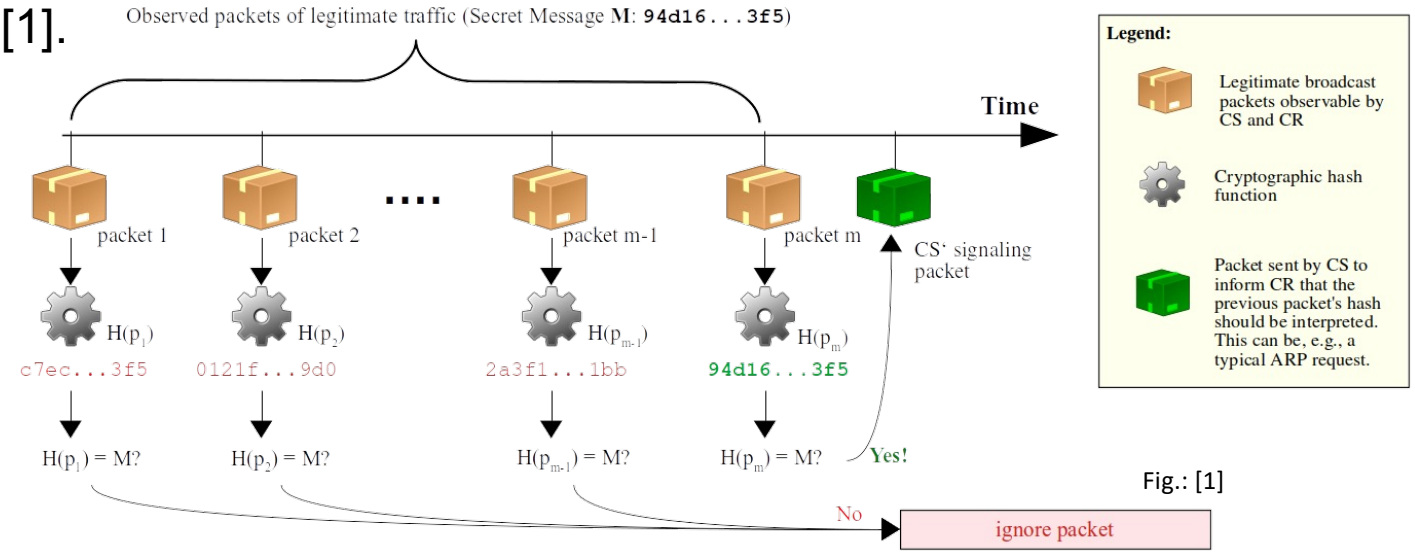
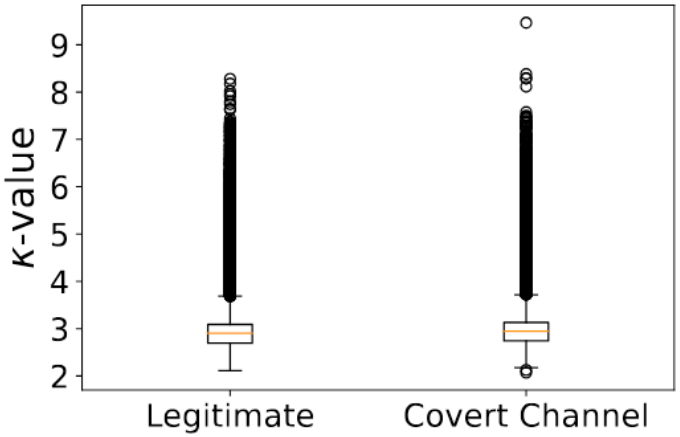


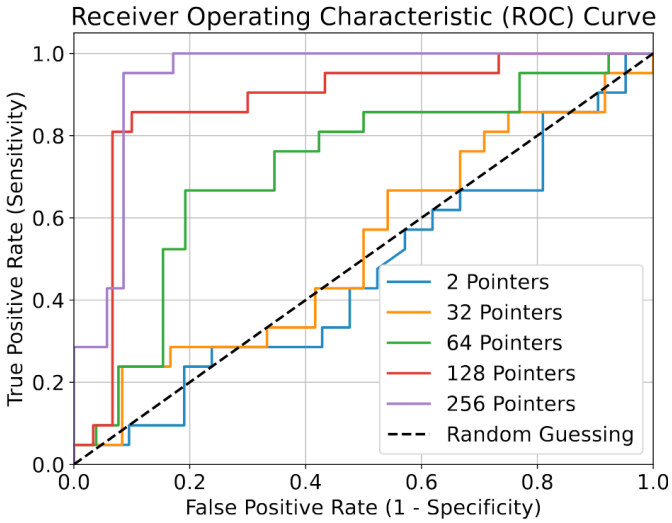
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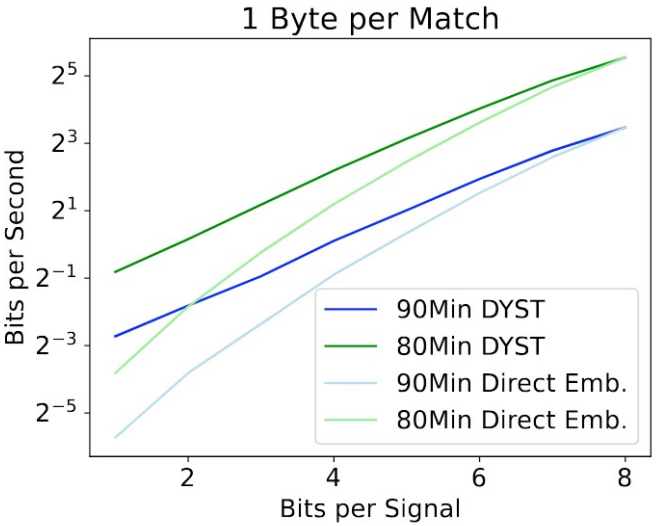
Types of (Network) Covert Channels: History Cov. Channels: DYST (Basic Version)

Detectability and Performance

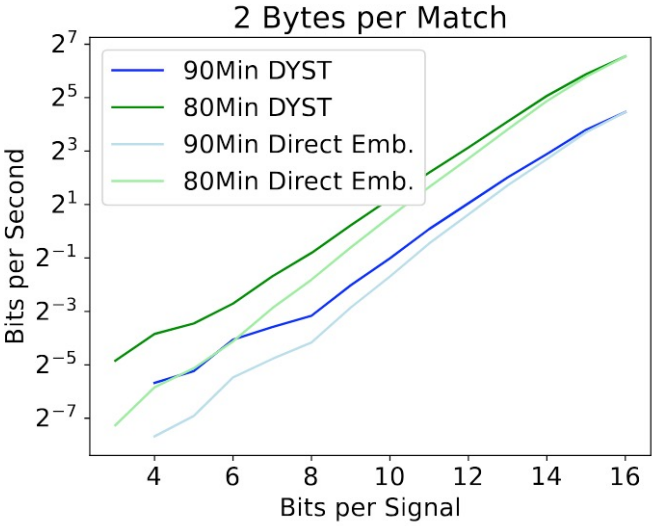




All Figures taken from [1]



(a) 1 byte/match



(b) 2 byte/match

[1] S. Wendzel, T. Schmidbauer, S. Zillien, J. Keller: DYST (Did You See That?): An Amplified Covert Channel That Points To Previously Seen Data. IEEE Transactions on Dependable and Secure Computing (TDSC), Vol. 22(1), 2025, DOI: 10.1109/TDSC.2024.3410679



History Covert Channels
pointing to textual data

OPPRESSION (published: ASIA CCS 2024)

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OPPRESSION (*Open-knowledge Compression*) [1]

- Still a **History Covert Channel**
- Still Covert Channel **Amplification**
- **But:**
  - does not point to recently observed network data
  - points to public Internet texts instead

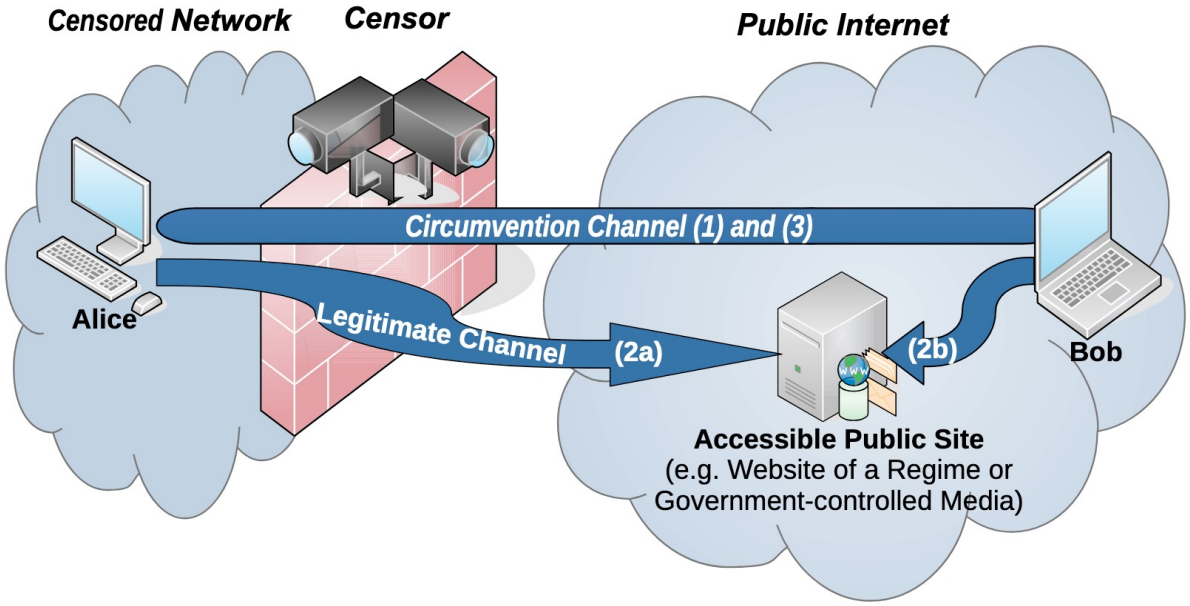


Figure 1: High-level Functioning of OPPRESSION

[1] S. Zillien, T. Schmidbauer, M. Kubek, J. Keller, S. Wendzel: Look What's There! Utilizing the Internet's Existing Data for Censorship Circumvention with OPPRESSION. ACM ASIA CCS, 2024. https://dl.acm.org/doi/pdf/10.1145/3634737.3637676

OPPRESSION (*Open-knowledge Compression*) [1]

- Sender and receiver both crawl public text repositories to create a local dictionary.

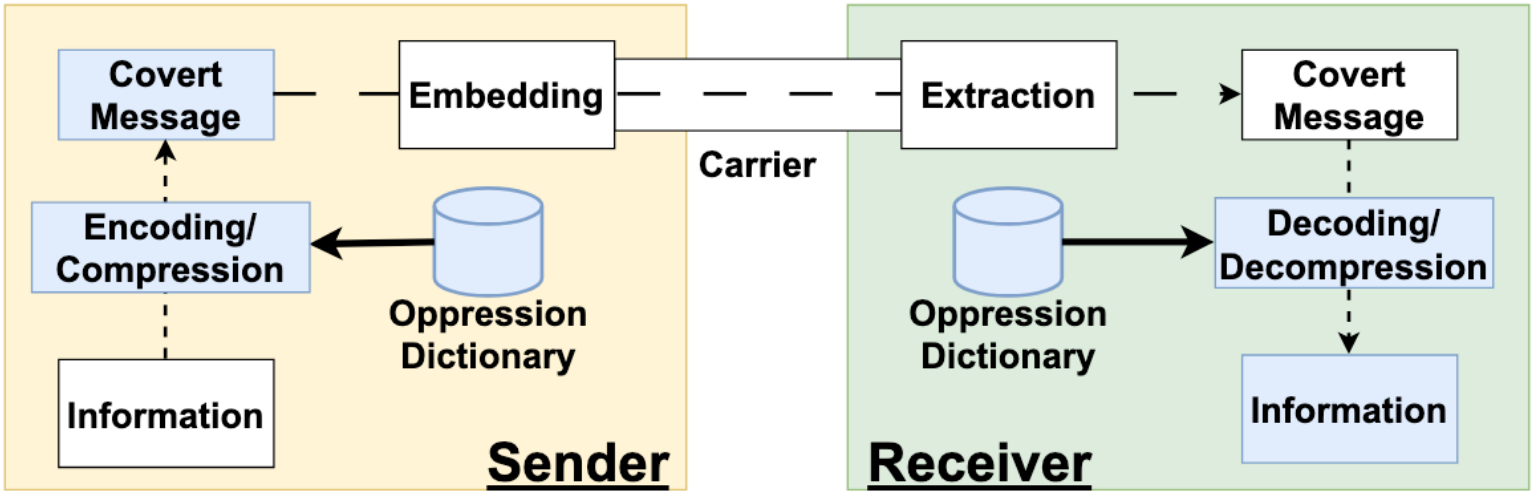


Figure 2: Contribution of OPPRESSION

[1] S. Zillien, T. Schmidbauer, M. Kubek, J. Keller, S. Wendzel: Look What's There! Utilizing the Internet's Existing Data for Censorship Circumvention with OPPRESSION. ACM ASIA CCS, 2024. https://dl.acm.org/doi/pdf/10.1145/3634737.3637676

OPPRESSION (*Open-knowledge Compression*) [1]

- In general: we are referring to a document (used to create a tree) and a particular node in that tree to represent a sentence.
- We can also spell words, if necessary (decreases compression).
- Two separate coding strategies (cf. paper)

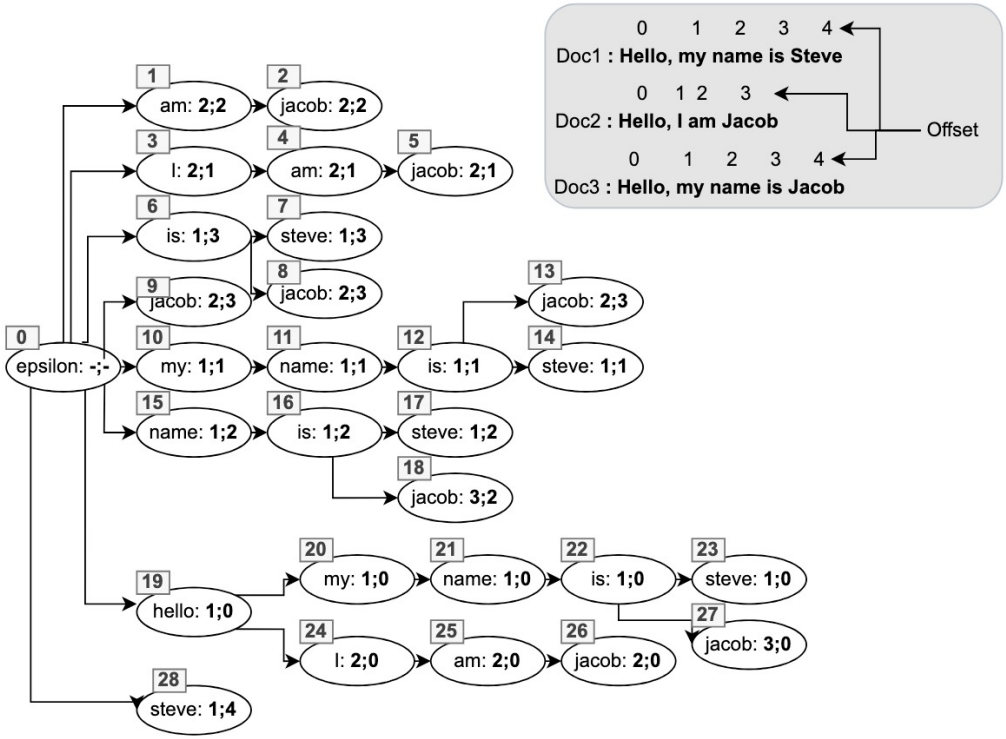


Figure 3: Document-pointer Tries

[1] S. Zillien, T. Schmidbauer, M. Kubek, J. Keller, S. Wendzel: Look What's There! Utilizing the Internet's Existing Data for Censorship Circumvention with OPPRESSION. ACM ASIA CCS, 2024. https://dl.acm.org/doi/pdf/10.1145/3634737.3637676

OPPRESSION (*Open-knowledge Compression*) [1]

- Exemplary results for one coding strategy and GZIP
- Also ran several more tests (see paper)!

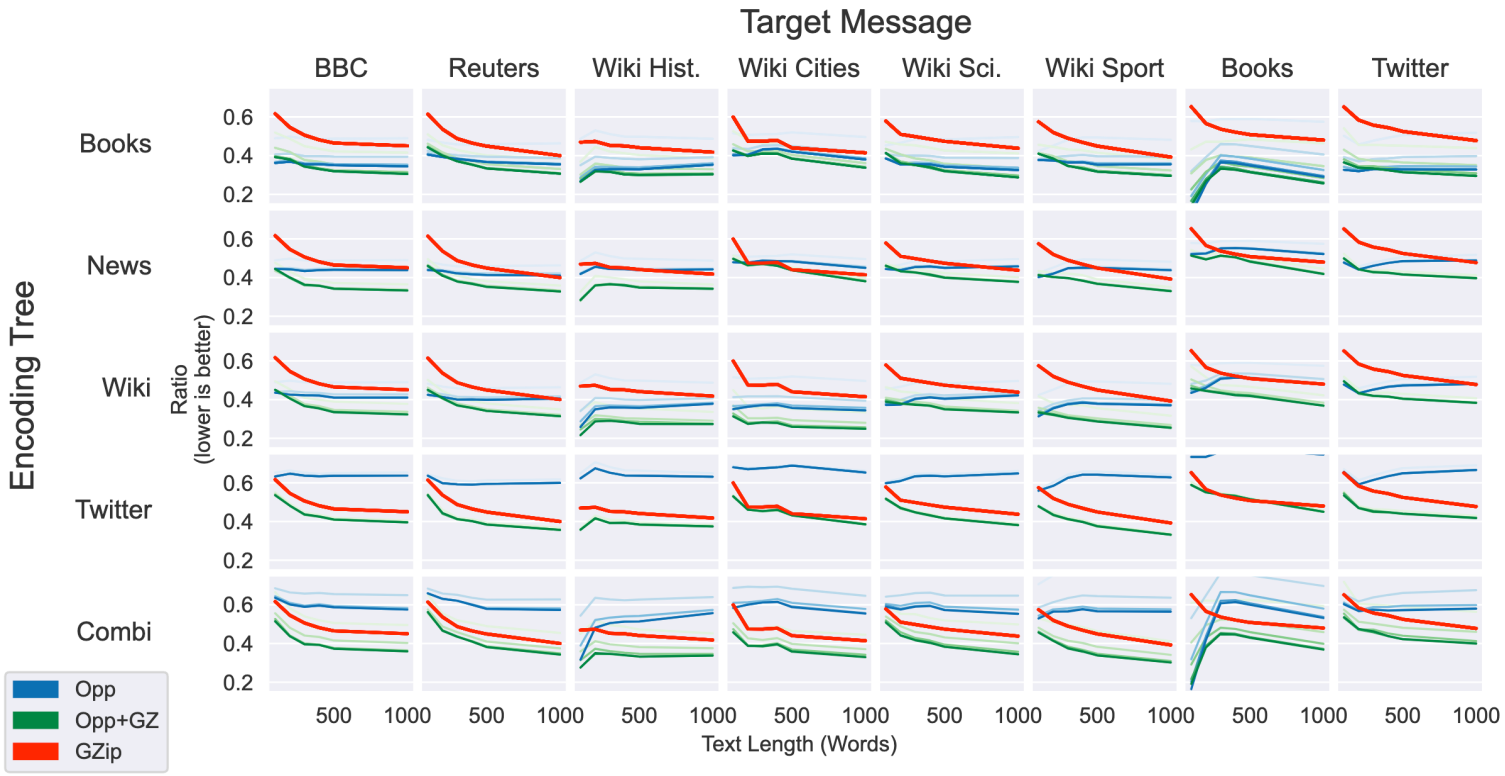


Figure 4: Evaluation Results: Doc. Pointer. Color shades indicate different tree depths (darker color signals a deeper tree).

[1] S. Zillien, T. Schmidbauer, M. Kubek, J. Keller, S. Wendzel: Look What's There! Utilizing the Internet's Existing Data for Censorship Circumvention with OPPRESSION. ACM ASIA CCS, 2024. https://dl.acm.org/doi/pdf/10.1145/3634737.3637676

OPPRESSION (*Open-knowledge Compression*) [1]

- Possible to enhance existing censorship circumvention tools:

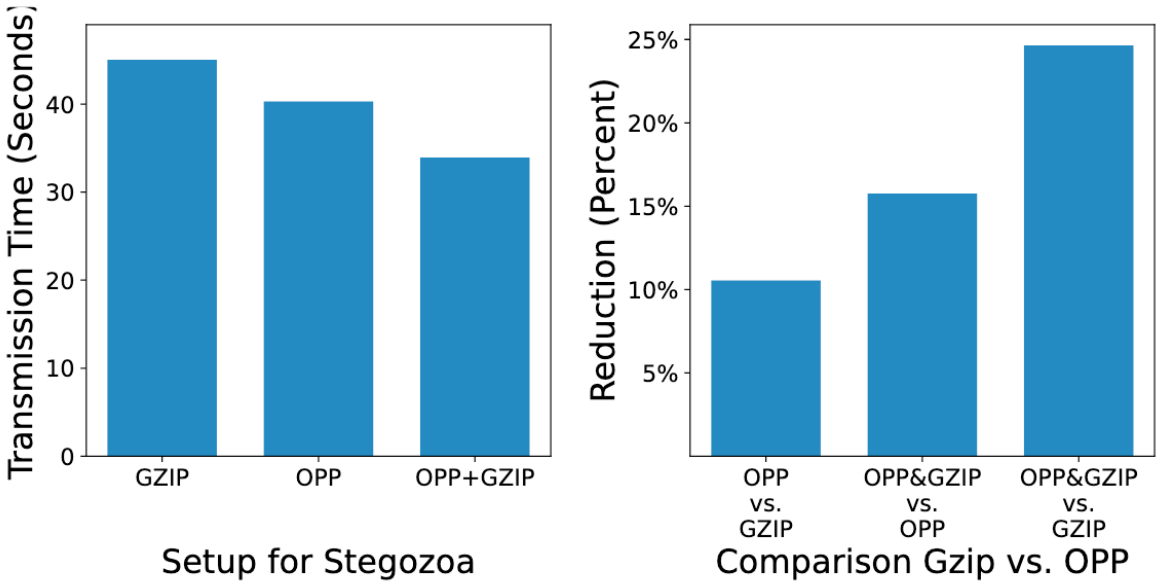
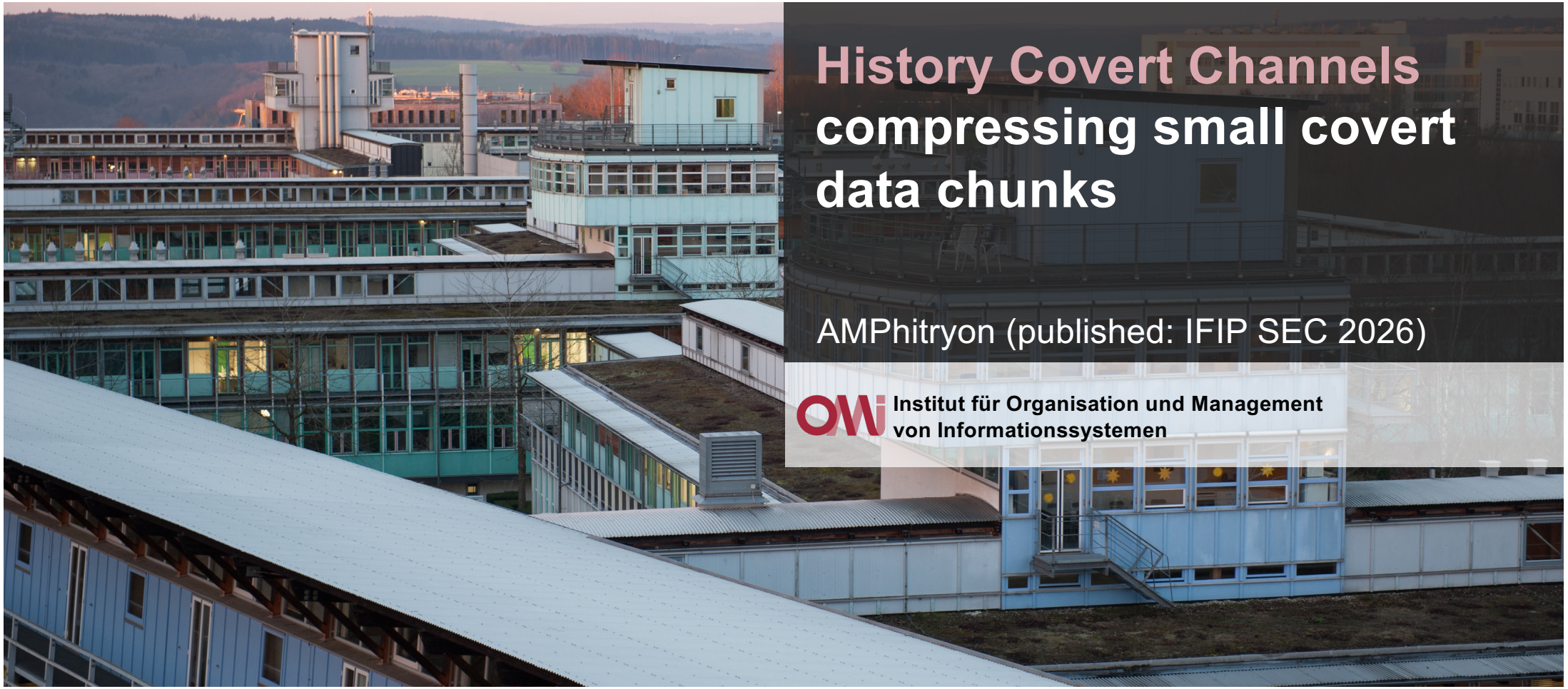


Figure 10: OPPRESSION with Stegozoa

[1] S. Zillien, T. Schmidbauer, M. Kubek, J. Keller, S. Wendzel: Look What's There! Utilizing the Internet's Existing Data for Censorship Circumvention with OPPRESSION. ACM ASIA CCS, 2024. https://dl.acm.org/doi/pdf/10.1145/3634737.3637676



History Covert Channels
compressing small covert data chunks

AMPhitryon (published: IFIP SEC 2026)

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AMPhitryon: Motivation

- Related work ignores small packets in censorship circumvention / covert channels
- But small packets leave traces too, so compression would be great to have!
- **But** state-of-the-art compressors not tailored to compress such small-sized data (<50 bytes)



Fig: Wikipedia / Detail of Amphitryon, from Infant Hercules strangling the serpents, from Herculaneum 'Basilica', Museo Archeologico, Naples, MN 9012

AMPhitryon

- Lossless compression method for arbitrary flows (especially covert channels / censorship circumvention flows)
- Can be integrated in censorship circumvention tools

… could also be used elsewhere, e.g. sensor data compression in IoT transmissions ☺

S. Wendzel, S. Zillien, S. Zander: AMPhitryon: Efficient small data compression for low-bandwidth covert channels, in Proc. IFIP SEC 2026, https://doi.org/10.1007/978-3-032-27993-4_32

Covert Channel Amplification Revisited

DYST
(IEEE TDSC 2025)

- Introduced Covert Channel Amplification
- Introduced History Covert Channels
- Limited transmission rate

OPPRESSION
(AsiaCCS 2024)

- Amplification only for textual data by referring to public online content
- Builds common dictionary for sender and receiver

Silent History Protocol (SHP)
(JISA 2026)

- Optimizes DYST's approach using relative pointers on flow characteristics
- Higher transmission rate than DYST

AMPhitryon
(IFIP SEC 2026)

- Creates flow-wise dictionaries to optimize amplification in context-specific manner (needs no crawling like OPPRESSION)
- For arbitrary censorship circumvention / covert channel tools

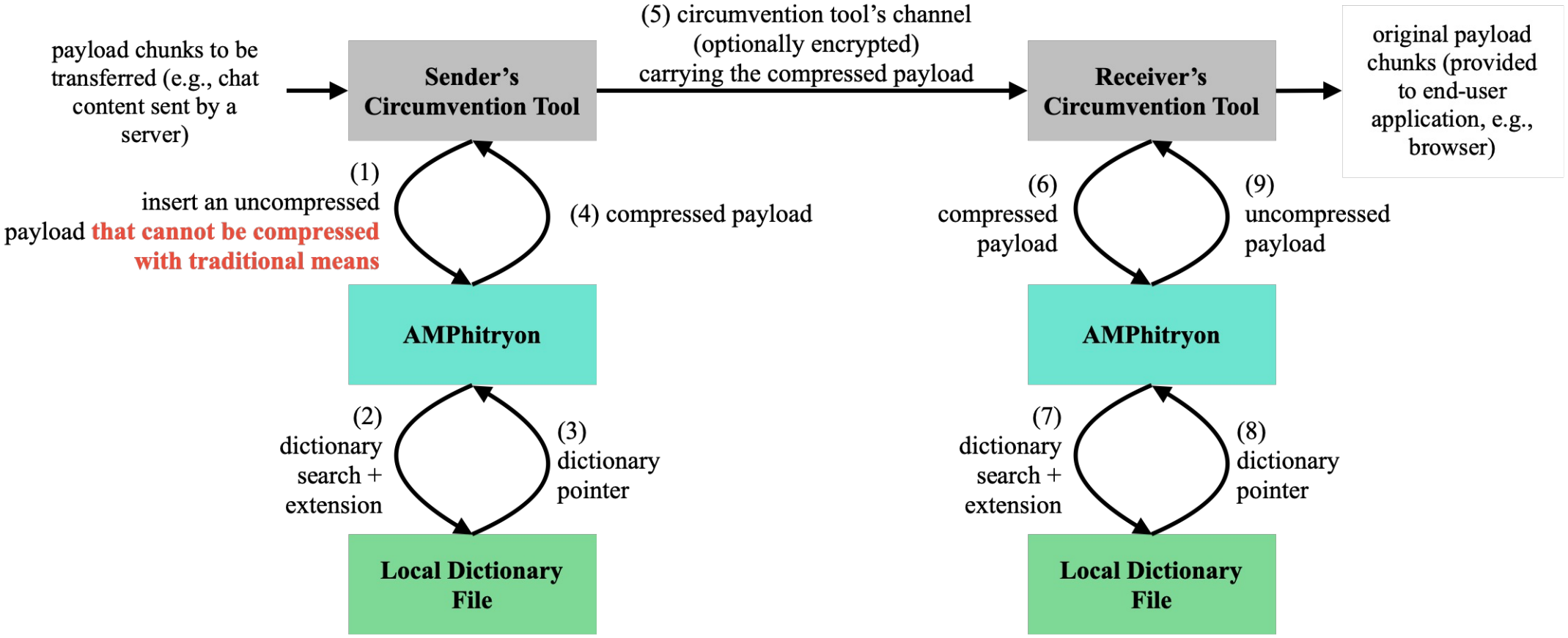
S. Wendzel, S. Zillien, S. Zander: AMPhitryon: Efficient small data compression for low-bandwidth covert channels, in Proc. IFIP SEC 2026, https://doi.org/10.1007/978-3-032-27993-4_32

AMPhitryon: Design

- Dictionary-based compression scheme with incremental dictionary
- Applied before encryption (sender) and after decryption (receiver)
- Communication-specific dictionaries are automatically built to optimally serve specific context
- Bootstrapping: compression becomes effective once chunk is found more than once (chunk size can be configured)
- Dictionary synchronization: majority of advanced covert channels provide reliability through control protocols
- Dictionary size: can be controlled with LRU policy
- Implementation: 500 LOC in C (including decompressor)

S. Wendzel, S. Zillien, S. Zander: AMPhitryon: Efficient small data compression for low-bandwidth covert channels, in Proc. IFIP SEC 2026, https://doi.org/10.1007/978-3-032-27993-4_32

AMPhitryon: How does it work?



S. Wendzel, S. Zillien, S. Zander: AMPhitryon: Efficient small data compression for low-bandwidth covert channels, in Proc. IFIP SEC 2026, https://doi.org/10.1007/978-3-032-27993-4_32

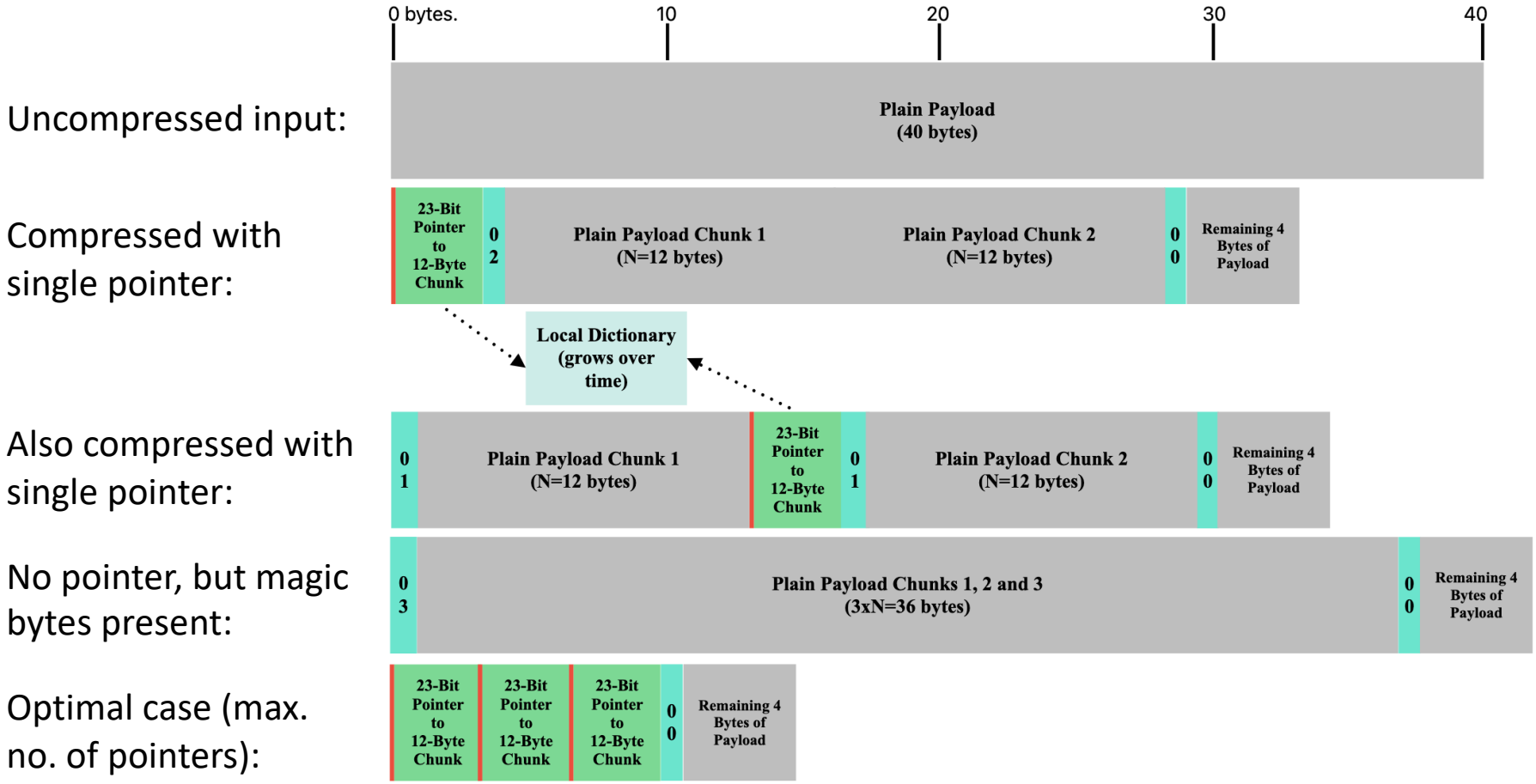
AMPhitryon: Compression format

- Amphitryon works with data split into fixed-size chunks of N bytes
- Sender and receiver add any unknown chunks to dictionary
- Sender replaces message chunk(s) already in dictionary with chunk pointer(s)
- Magic bit that indicates whether following 23 bits are a chunk pointer or following 7 bits are number of data chunks following
  - Number of 0 chunks indicates final (potentially incomplete) chunk



S. Wendzel, S. Zillien, S. Zander: AMPhitryon: Efficient small data compression for low-bandwidth covert channels, in Proc. IFIP SEC 2026, https://doi.org/10.1007/978-3-032-27993-4_32

AMPhitryon: Compression format examples



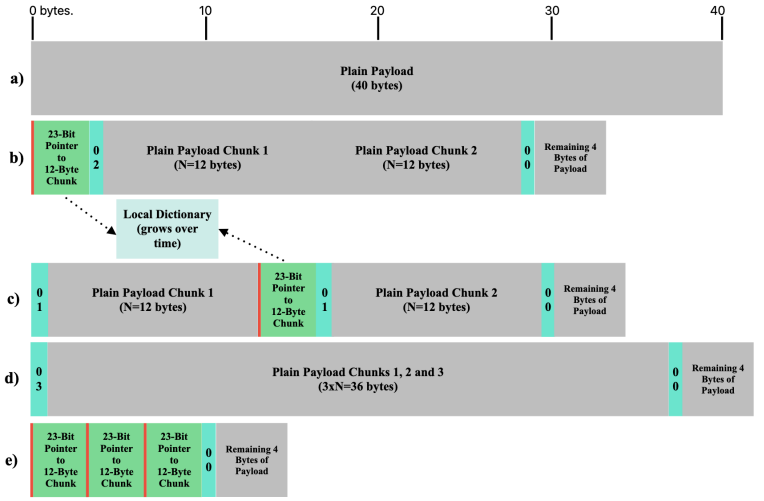
S. Wendzel, S. Zillien, S. Zander: AMPhitryon: Efficient small data compression for low-bandwidth covert channels, in Proc. IFIP SEC 2026, https://doi.org/10.1007/978-3-032-27993-4_32

AMPhitryon: Compressed size

Compressed size C of original uncompressed message O is within

$$3P \leq C \leq 3P + (O - N\,P) \cdot ((N + 1)/N),$$

with P being number of 3-byte pointers and N being chunk size



S. Wendzel, S. Zillien, S. Zander: AMPhitryon: Efficient small data compression for low-bandwidth covert channels, in Proc. IFIP SEC 2026, https://doi.org/10.1007/978-3-032-27993-4_32

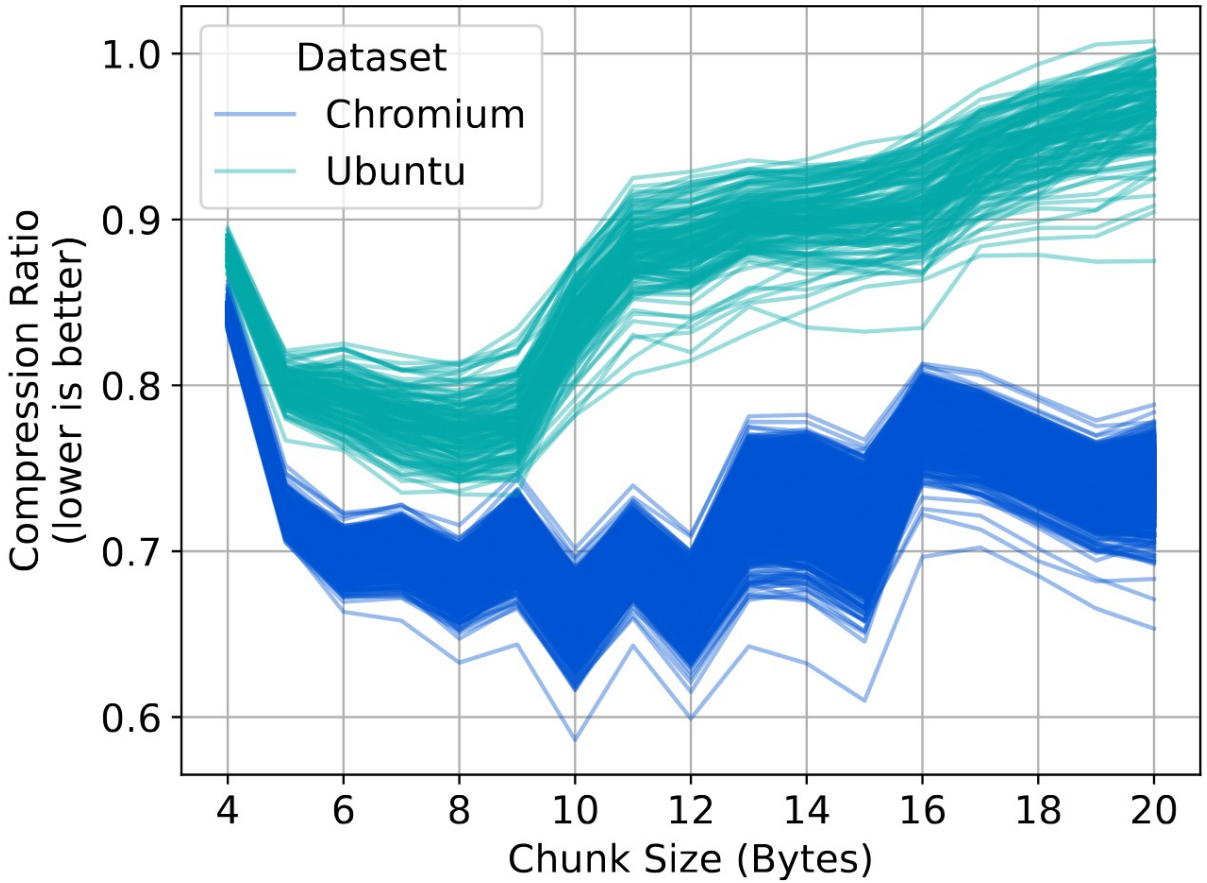
Evaluation: Datasets

- Used publicly available chat protocols to simulate covert channel-based chat
  - **Ubuntu**: https://doi.org/10.18653/v1/P19-1374
  - **Chromium**: https://doi.org/10.5281/zenodo.2590548
- Filtered out long messages (> 50 bytes) and outliers (e.g. messages with many consecutive instances of same character)

Dataset	Time-Frame	Messages	Log Files
Ubuntu	2005-2018	220,210	153
Chromium	2008-2016	2,853,885	1,357

S. Wendzel, S. Zillien, S. Zander: AMPhitryon: Efficient small data compression for low-bandwidth covert channels, in Proc. IFIP SEC 2026, https://doi.org/10.1007/978-3-032-27993-4_32

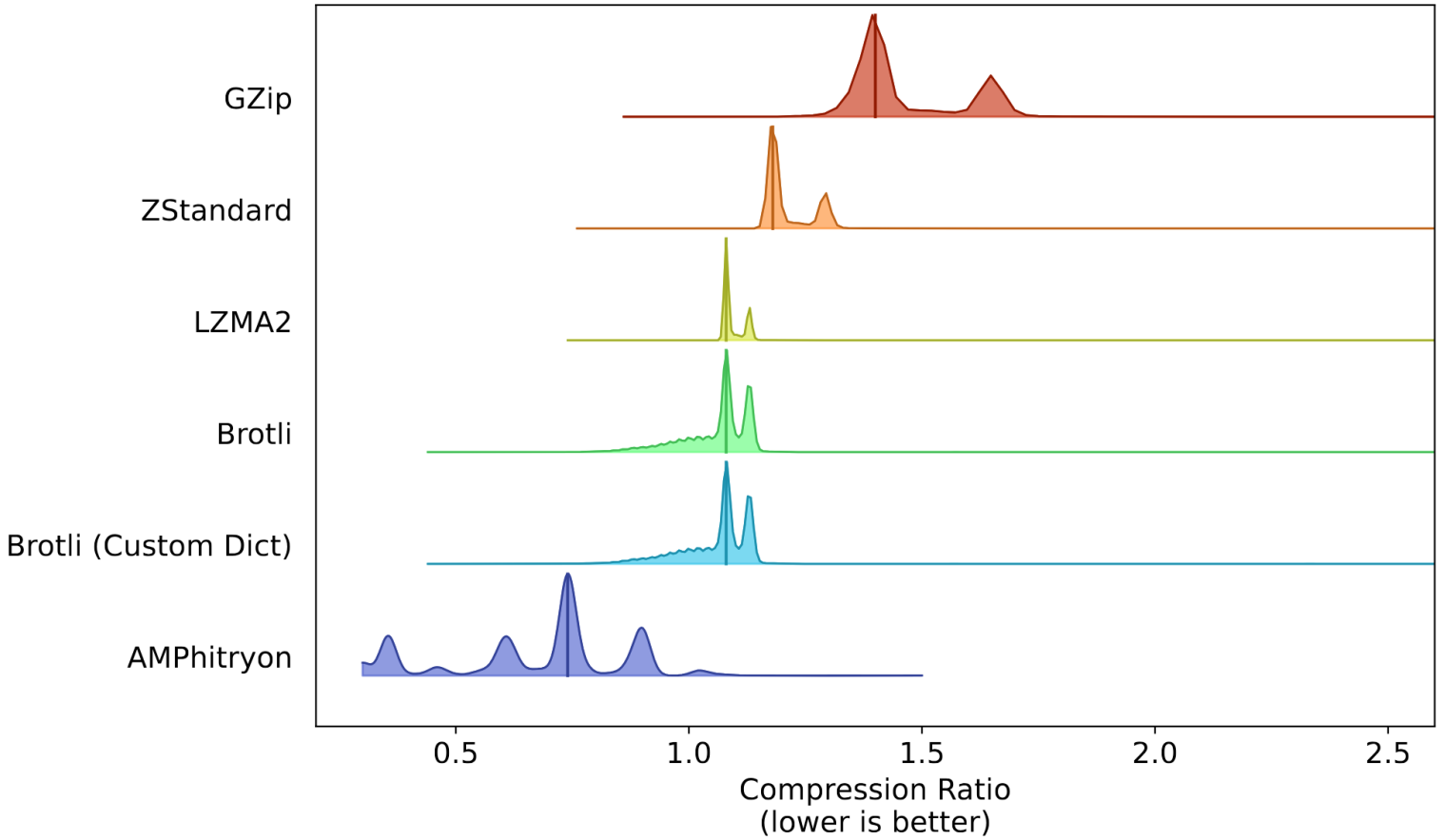
Evaluation: Compression performance by chunk size



- Best performance with chunk size between 7 and 12 bytes

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Evaluation: Comparison with state-of-the-art compressors



AMPhitryon outperforms other compressors!

Figures shows results for chunk size=10

S. Wendzel, S. Zillien, S. Zander: AMPhitryon: Efficient small data compression for low-bandwidth covert channels, in Proc. IFIP SEC 2026, https://doi.org/10.1007/978-3-032-27993-4_32

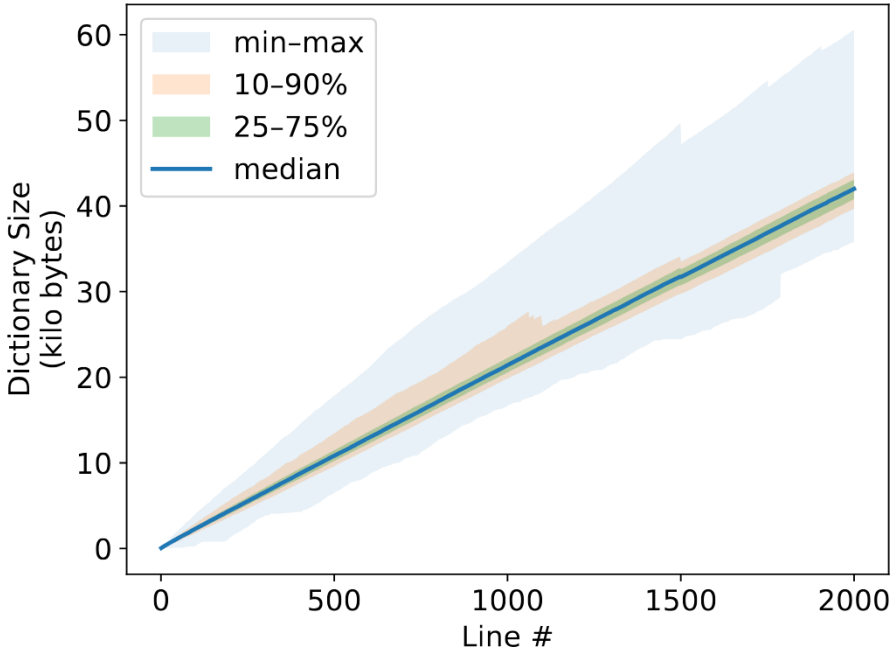
Evaluation: Comparison with state-of-the-art compressors

- Minimum, maximum, mean and median compression rates using chunk sizes 4 to 20

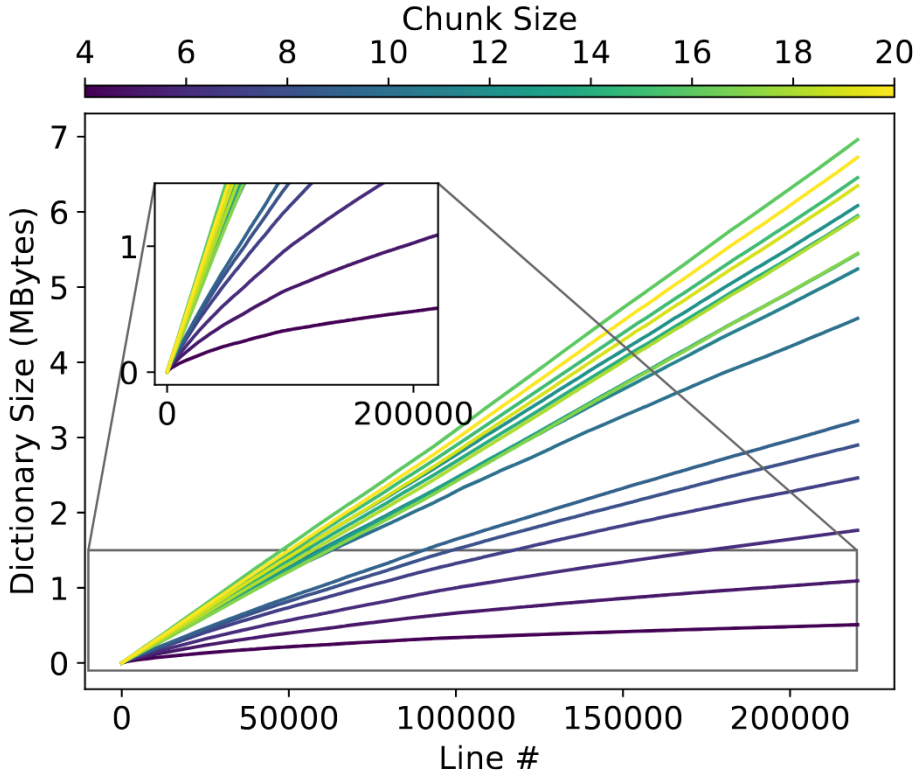
Algorithm	Min	Max	Mean	Median
AMPhitryon	0.30	1.50	0.67	0.70
Gzip	0.86	11.0	1.47	1.40
LZMA2	0.74	3.0	1.10	1.08
Brotli	0.44	3.0	1.06	1.08
Brotli (Custom Dictionary)	0.44	3.0	1.06	1.08
ZStandard	0.76	5.5	1.22	1.18

S. Wendzel, S. Zillien, S. Zander: AMPhitryon: Efficient small data compression for low-bandwidth covert channels, in Proc. IFIP SEC 2026, https://doi.org/10.1007/978-3-032-27993-4_32

Evaluation: Dictionary growth over time



Average dictionary size by chat line (chunk size 10)

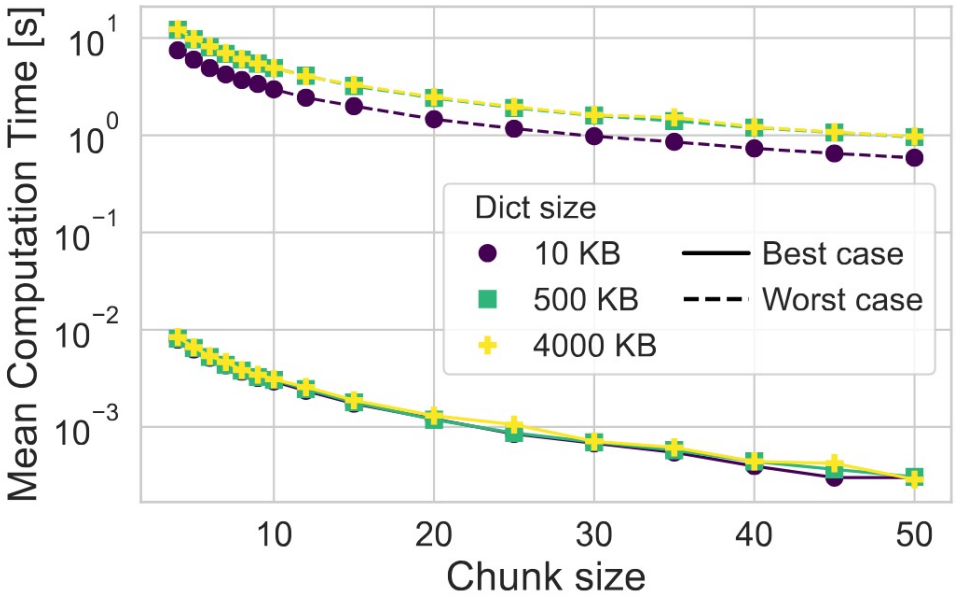


Dictionary size by chat line (large file)

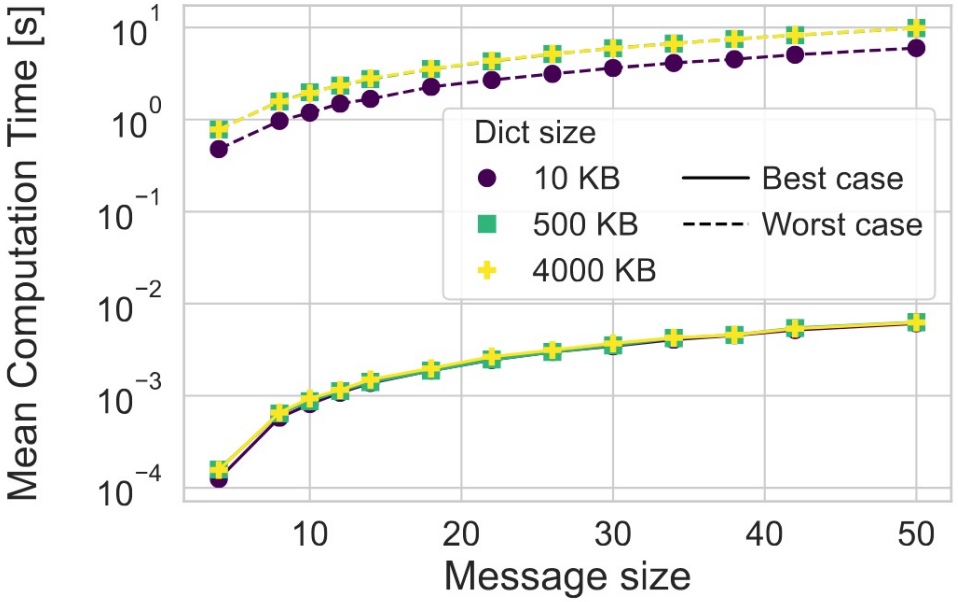
S. Wendzel, S. Zillien, S. Zander: AMPhitryon: Efficient small data compression for low-bandwidth covert channels, in Proc. IFIP SEC 2026, https://doi.org/10.1007/978-3-032-27993-4_32

Evaluation: Computation time

Rewrite in progress – eliminating file I/O and processing everything in memory



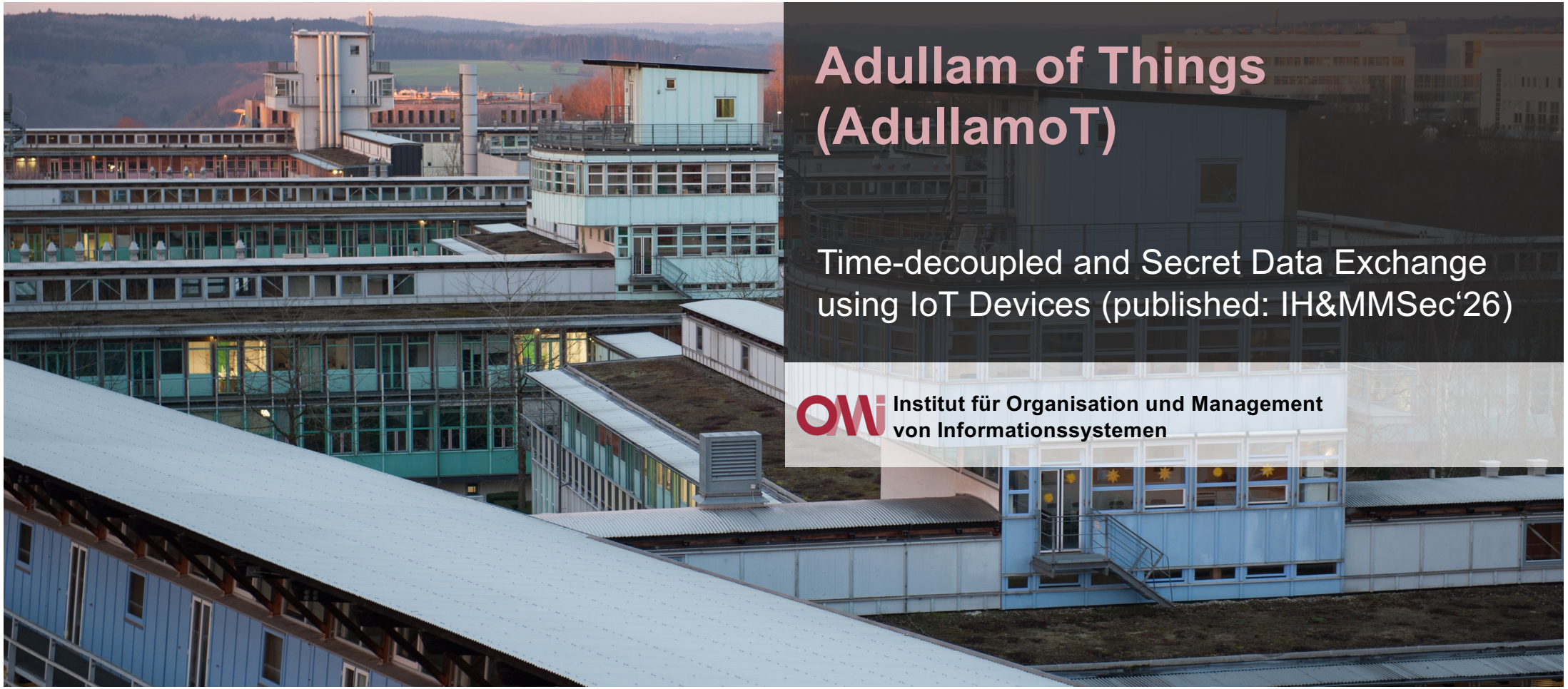
Computation time vs chunk size



Computation time vs message size

S. Wendzel, S. Zillien, S. Zander: AMPhitryon: Efficient small data compression for low-bandwidth covert channels, in Proc. IFIP SEC 2026, https://doi.org/10.1007/978-3-032-27993-4_32

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Adullam of Things (AdullamoT)

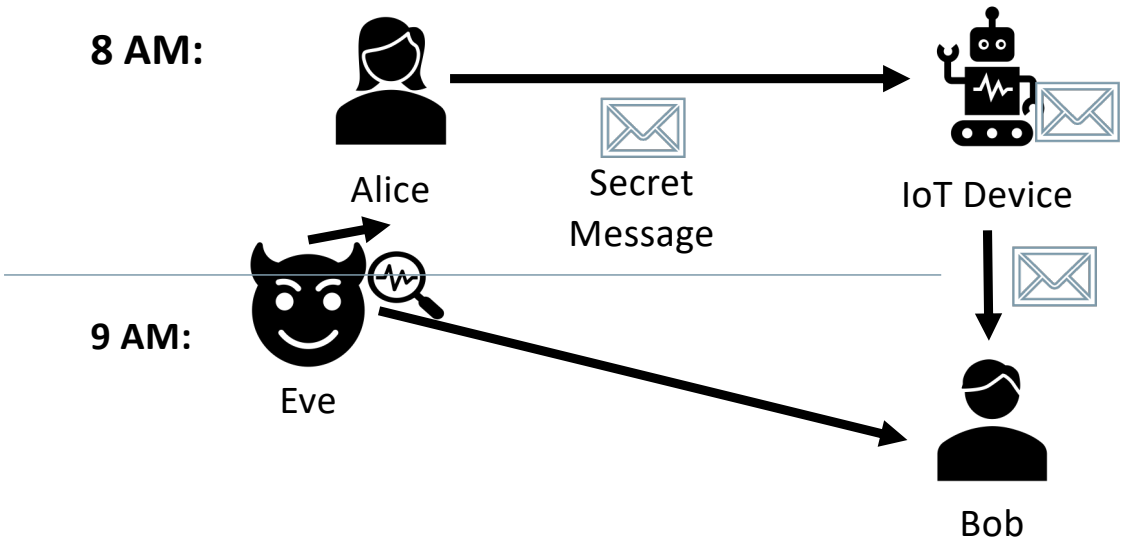
Time-decoupled and Secret Data Exchange using IoT Devices (published: IH&MMSec‘26)

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Threat Scenario 1: Alice and Bob do not want to reveal that they exchange secret information

Whistlebower **Alice** and journalist **Bob** utilize **IoT devices** to **relay secret information**.

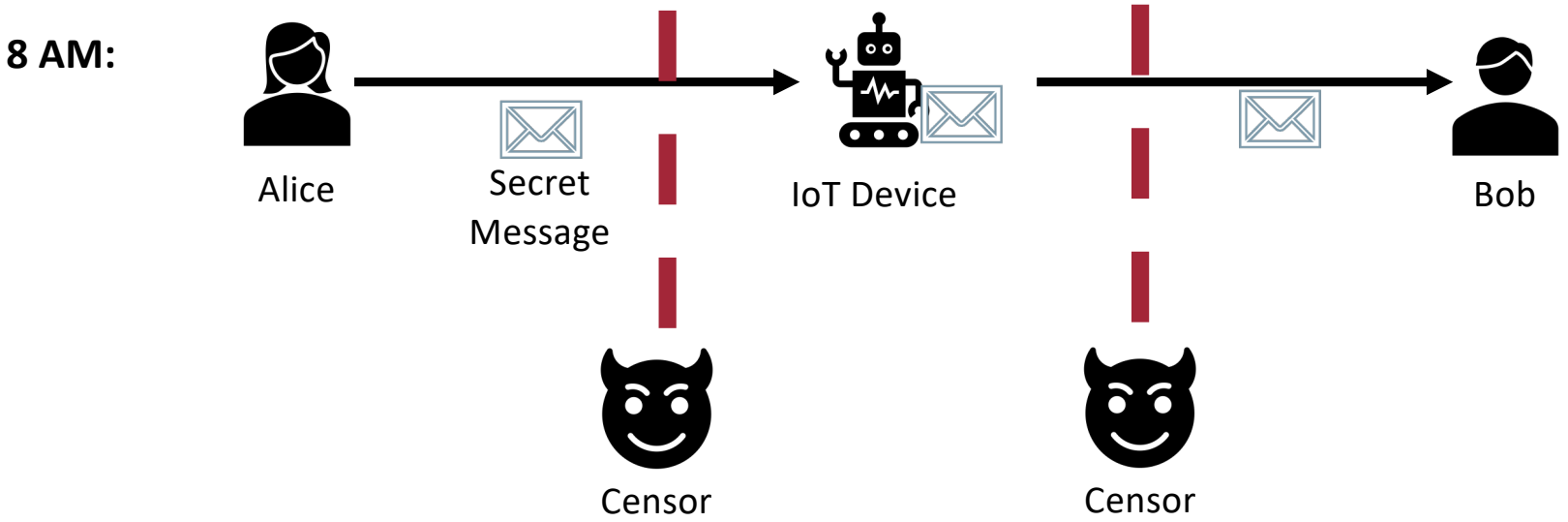
Eve tries to determine whether Alice and Bob exchange secret information.



S. Wendzel: AdullamoT: Using IoT Devices as Relays for Time-decoupled Secret Exchange & Censorship Circumvention, in Proc. IH&MMSec 2026, ACM, 10.1145/3785353.3815068

Threat Scenario 2: Alice and Bob want to bypass censorship – stealthiness not a goal

High transmission rate, max. number of messages/sec to be exchanged.



S. Wendzel: AdullamoT: Using IoT Devices as Relays for Time-decoupled Secret Exchange & Censorship Circumvention, in Proc. IH&MMSec 2026, ACM, 10.1145/3785353.3815068

AdullamoT approach

Using **IoT devices** for covert data storage by slightly **altering attributes (cover objects) of these devices**.

Requirements:

1. The modified attributes on the IoT devices should be accessible by COTS network devices without a special key/credential or software.
2. Modifications of the attributes should not be directly recognizable by end-users.

Optional:

- Using devices that are already present, preferably in an ”idle” mode.

S. Wendzel: AdullamoT: Using IoT Devices as Relays for Time-decoupled Secret Exchange & Censorship Circumvention, in Proc. IH&MMSec 2026, ACM, 10.1145/3785353.3815068

IoT Device Selection



Brother HL-L2375DW



HP LaserJet Pro M148dw



HP LaserJet M15w



Block SB100

S. Wendzel: AdullamoT: Using IoT Devices as Relays for Time-decoupled Secret Exchange & Censorship Circumvention, in Proc. IH&MMSec 2026, ACM, 10.1145/3785353.3815068

IoT Device Selection



Brother HL-L2375DW



HP LaserJet Pro M148dw



HP LaserJet M15w

Change printer “location” using web-interfaces (was unprotected in case of all three printers although being used for several years!).

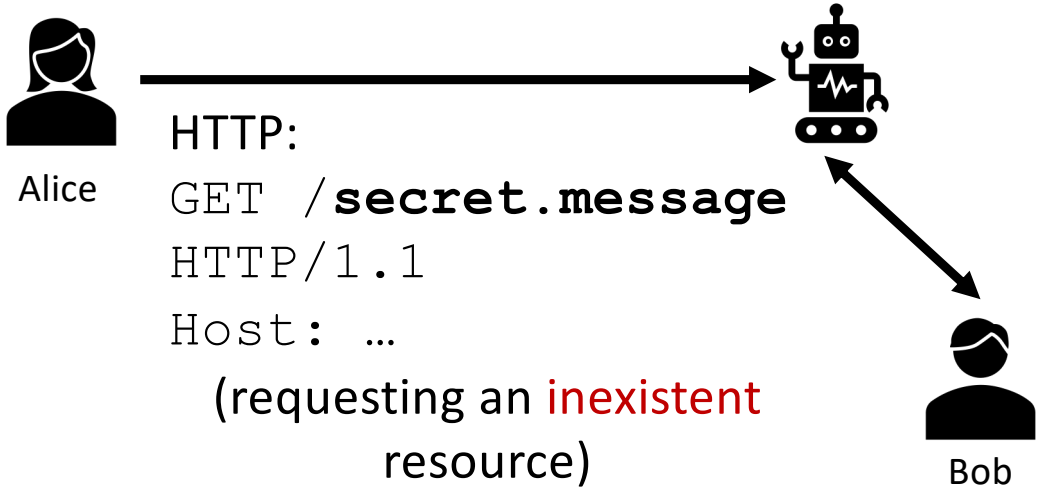
- Scripts to perform HTTP(S)-based POST command for altering settings and GET commands. –> Reversible by Bob!
- Brother printer required handling of **CSRF tokens**.

S. Wendzel: AdullamoT: Using IoT Devices as Relays for Time-decoupled Secret Exchange & Censorship Circumvention, in Proc. IH&MMSec 2026, ACM, 10.1145/3785353.3815068

IoT Device Selection



Block SB100



TCP port 514 (logging):

```
%(Thread2): [151855.884631]
FSFS    (2):
fsfsFlashFileHandleOpen: File
'flash://secret.message' not
found
%(Thread2): [151855.886162]
WFSAPI (2): File not found
```

S. Wendzel: AdullamoT: Using IoT Devices as Relays for Time-decoupled Secret Exchange & Censorship Circumvention, in Proc. IH&MMSec 2026, ACM, 10.1145/3785353.3815068

IoT Device Selection



Brother HL-L2375DW



HP LaserJet Pro M148dw



HP LaserJet M15w



Block SB100

Manufacturer, Model	Type	Connectivity
Brother HL-L2375DW	laser printer	WiFi
HP LaserJet Pro M148dw	laser printer	Ethernet
HP LaserJet M15w	laser printer	WiFi
Block SB100	smart speaker	Ethernet

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IoT Device Selection



Brother HL-L2375DW



HP LaserJet Pro M148dw



HP LaserJet M15w



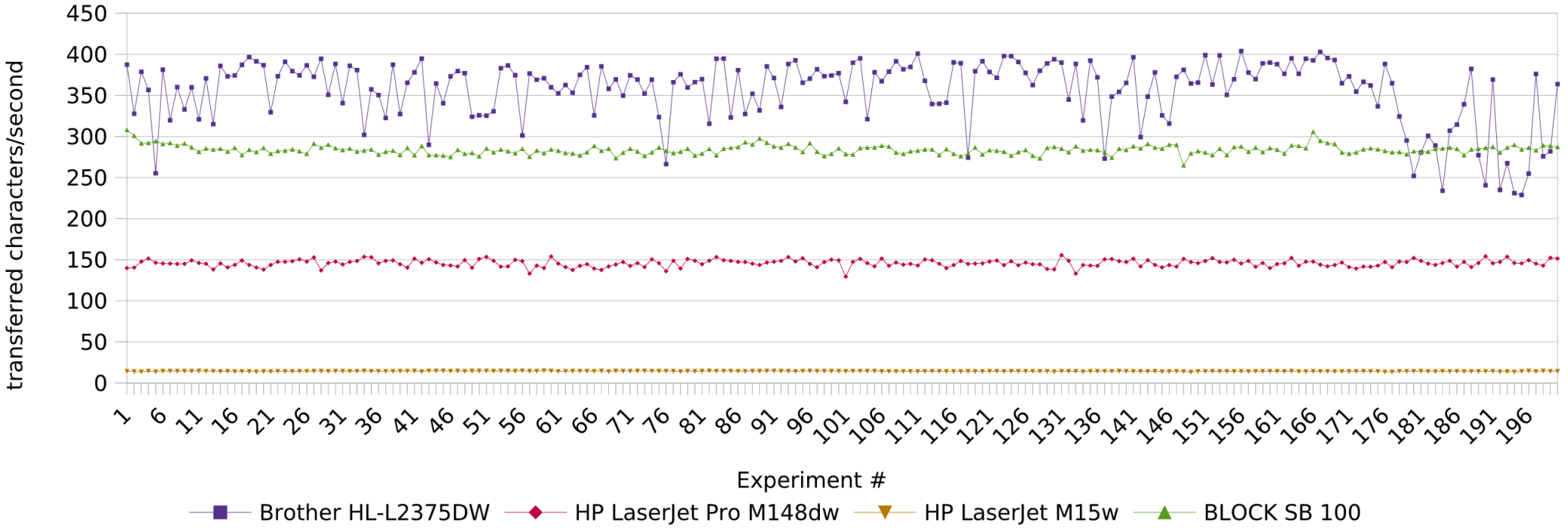
Block SB100

Manufacturer, Model	Type	Connectivity	Implemented Protocol	$s_{max}(d_i)$	Mode
Brother HL-L2375DW	laser printer	WiFi	send + wait for simple ACK	100 characters	turned on, inactive
HP LaserJet Pro M148dw	laser printer	Ethernet	send + wait for simple ACK	255 characters	turned on, inactive
HP LaserJet M15w	laser printer	WiFi	send + wait for simple ACK	64 characters	turned on, inactive
Block SB100	smart speaker	Ethernet	blind sending	59 characters	standby mode

S. Wendzel: AdullamoT: Using IoT Devices as Relays for Time-decoupled Secret Exchange & Censorship Circumvention, in Proc. IH&MMSec 2026, ACM, 10.1145/3785353.3815068

Evaluation: Transmission Rate

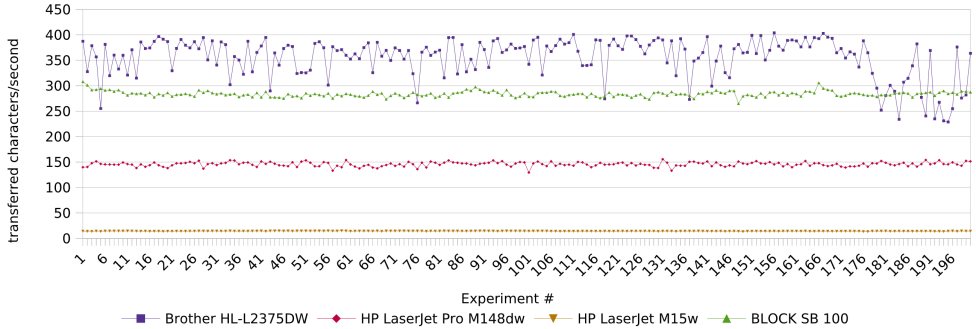
- Conducted ≥ 200 measurements per device, embedded max. possible ASCII characters per message (6,400 - 10,000 characters transferred per data point).



S. Wendzel: AdullamoT: Using IoT Devices as Relays for Time-decoupled Secret Exchange & Censorship Circumvention, in Proc. IH&MMSec 2026, ACM, 10.1145/3785353.3815068

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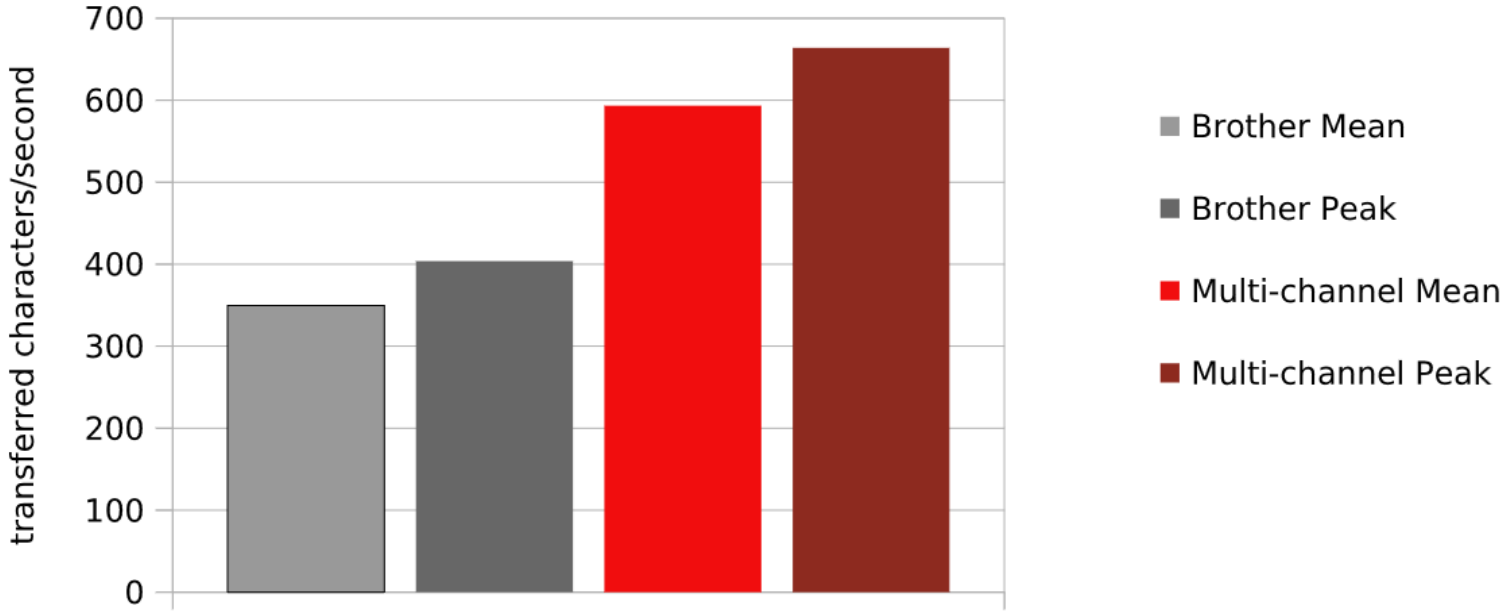
Manufacturer, Model	Min.	Max.	Mean	σ
Brother HL-L2375DW	202.72	403.98	349.62	45.45
HP LaserJet Pro M148dw	129.32	155.61	145.51	4.26
HP LaserJet M15w	13.34	14.90	14.23	0.24
Block SB100	264.81	308.17	283.85	5.39

Both, best and worst performance, reached via **WiFi**

S. Wendzel: AdullamoT: Using IoT Devices as Relays for Time-decoupled Secret Exchange & Censorship Circumvention, in Proc. IH&MMSec 2026, ACM, 10.1145/3785353.3815068

Evaluation: Transmission Rate (Multi-Channel Mode)

- Multi-channel mode, optimized $f = \sum_1^n p_i \cdot s_{max}(d_i)$, threshold c=.2, $0 < c \leq p_i \leq 1$ for throughput (see chapter 6), using 3 devices (p=.421 Brother printer, p=.359 Block smart speaker, and p=.22 HP LaserJet Pro).



S. Wendzel: AdullamoT: Using IoT Devices as Relays for Time-decoupled Secret Exchange & Censorship Circumvention, in Proc. IH&MMSec 2026, ACM, 10.1145/3785353.3815068

Evaluation: Robustness

Immediate Retrieval after $\leq$ 10 sec:

- 0% measured errors for **all** devices, even with max. transmission rate

Time-decoupled Relaying:

- Unlimited **error-free** relay for all **printers** (unless another user would change the printer's location setting).
- Smart **speaker** limited due to log cache limit (N=16/2); log messages overwrite those caused by Alice. However, storage duration of one hour >92% retrieval rate if only 50% of capacity is used.

Examples:

```
(Thread2): [ 506731.853883] SPOTIFY (2):
spotifyTspTaskEntry(): cmd:4 error:0
(Thread2): [ 582271.514119] SPOTIFY (2):
spotifyTspTaskEntry(): cmd:-1335047288 error:0
```

S. Wendzel: AdullamoT: Using IoT Devices as Relays for Time-decoupled Secret Exchange & Censorship Circumvention, in Proc. IH&MMSec 2026, ACM, 10.1145/3785353.3815068

Evaluation: Detectability

- **Network-level:**
  - Trivial, in case an anomaly detection monitors the altered attributes on printers or applies a threshold $n \geq 3$ established TCP connections/sec. Rarely the case in a real-world office/café network setting.

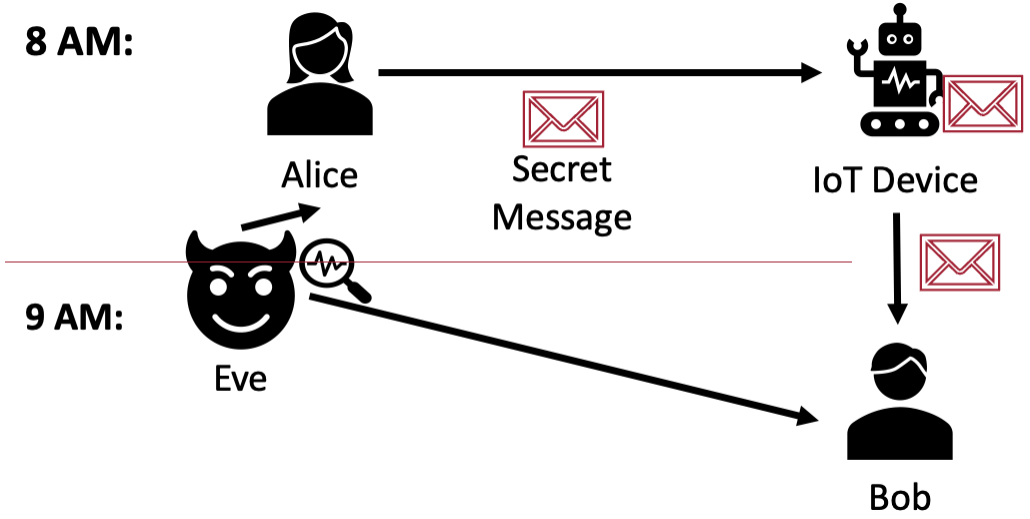
- **Audio-Visual Detectability:**
  - Neither the printers nor the smart speaker had to be "used" (e.g., commanded to print papers or to play sound, respectively). All devices were kept inactive or in standby mode.
  - No audible or visual signal (blinking LEDs or changes on displays) was generated by the covert communication that could be interpreted as an anomaly.

- **Congestion-based Detectability:**
  - Observed delays of up to 10 sec when printers were used with max. transmission rate.
  - Observed no (load-caused) restarts.

S. Wendzel: AdullamoT: Using IoT Devices as Relays for Time-decoupled Secret Exchange & Censorship Circumvention, in Proc. IH&MMSec 2026, ACM, 10.1145/3785353.3815068

Problem of Limited Storage Space and Low Relay Frequency

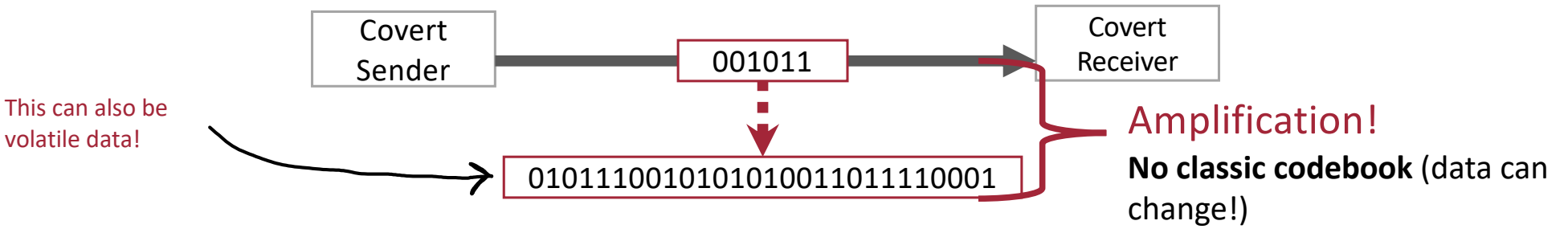
- Let us go back to the scenario where Alice and Bob **de-couple** their exchanges -> they will transfer **only few messages, but their size is also limited by** s_{max}.
- Thus, they must exchange as much data as possible **per message**.



S. Wendzel: AdullamoT: Using IoT Devices as Relays for Time-decoupled Secret Exchange & Censorship Circumvention, in Proc. IH&MMSec 2026, ACM, 10.1145/3785353.3815068

Solution: Covert (Message Size) Amplification*

- If we have non-random data, e.g., text messages, we can apply **covert channel amplification** [1], i.e., amplify the message size using pointers:
- Procedure:
  - Replace secret message M with concatenated pointers $p_1 || \ldots || p_n$ referring to message fragments $m_1, \ldots, m_n$ found elsewhere, while $\sum|p_i| \leq s_{max}(d_i)$.
    - Make sure that $|m_i| \geq |p_i|$ and $\sum|m_i| > \sum|p_i|$ -> amplification of message size!



*** Did not fit into the paper -> electronic supplement on repository linked in the paper.**

[1] S. Wendzel et al.: ***DYST*** *(Did you see that?): An amplified covert channel that points to previously seen data,* in: IEEE Trans. Dependable and Secure Computing (TDSC), 2025.

Discussion and Limitations

Discussion:

- Using public devices on the Internet probably feasible but ethical limitations
  - `shodan.io` returned 1,160 results for “HP LaserJet Pro”, 4,306 results for “HP LaserJet”, none for the Block speaker.
- Countermeasure: protecting/disabling management interfaces (not smart speaker!)
- Reversibility can be done by Bob in case of the printers.

Limitations:

- Small number of only 4 devices (not able to generalize our results)
- Evaluated no Internet-based / real-world censorship circumvention setting (testbed in local network)

S. Wendzel: AdullamoT: Using IoT Devices as Relays for Time-decoupled Secret Exchange & Censorship Circumvention, in Proc. IH&MMSec 2026, ACM, 10.1145/3785353.3815068

Summary of AdullamoT

- **AdullamoT**, a method for **covertly relaying data in a (time-decoupled) manner** while **utilizing IoT devices to store and exchange secret data**.
- Proof-of-concept implementation available.
- Evaluated essential performance characteristics using four common commercial off-the-shelf (COTS) IoT devices.
- Our approach could be improved and enhanced to aid **real-world censorship circumvention** as well as by applying covert ch. **amplification**.

S. Wendzel: AdullamoT: Using IoT Devices as Relays for Time-decoupled Secret Exchange & Censorship Circumvention, in Proc. IH&MMSec 2026, ACM, 10.1145/3785353.3815068

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First Aspects of Covert Channel Measurement

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How to „measure“ covert channels?

- Introduction of **Covertness** by Giani et al. [1]:

  Covertness ∝ (Capacity of the medium – Transmission Rate)

  If the whole capacity of a transmission medium (e.g. network packets or an audio CD) is used, the covertness is zero, leading to a trivial detection. However, if only a tiny fraction of the capacity is used, the covertness can remain close to one.

[1] A. Giani, V. H. Berk, G. V. Cybenko: Data Exfiltration and Covert Channels, Sensors, and Command, Control, Communications, and Intelligence (C3I) Technologies for Homeland Security and Homeland Defense V. Vol. 6201. International Society for Optics and Photonics, 2006.

Let’s have a look at Fig. 1 and 2 of their paper.

How to „measure“ covert channels?

- **Steganographic Cost (SC)** by Mazurczyk et al. [1]:
  - Measure of degradation or distortion of a carrier caused by the application of a steganographic method.
  - Calculation depends on context. For instance, for *LACK* steganography, which exploits packet loss, the **SC** can be calculated using the *Mean Opinion Score* (MOS) as a **difference in quality of the voice signal** without (**RQ**) and with LACK applied (**LQ**):

$$SC_{T-LACK}(t) = \Delta MOS(t) = RQ(t) - LQ(t)$$

  - For *Retransmission Steganography* (RSTEG), one can calculate the **retransmission difference** R_D instead:

$$SC_{T-RSTEG} = R_D = R_{N-RSTEG} - R_N$$

    - $R_{N-RSTEG}$ denotes retransmissions in the network with RSTEG and R_N the network's retransmissions without applying RSTEG.

[1] W. Mazurczyk, S. Wendzel, I. Azagra Villares, K. Szczypiorski: On importance of steganographic cost for network steganography, SCN, 9(8), 781-790, Wiley, 2016.

How to „measure“ covert channels?

Covert Channel Bandwidth [1]:

$$B = b \cdot (T_R + T_S + 2T_{CS})^{-1}$$

b: Amount of transferred information per message.

T_R: Time needed to receive a covert message

T_s: Time to send a covert message

T_{CS}: Time equired for the context switch between processes

[1] C-R. Tsai and V. D. Gligor, “A bandwidth computation model for covert storage channels and its applications,” in Proc. IEEE Symp. on Security and Privacy, 1988, pp. 108–121